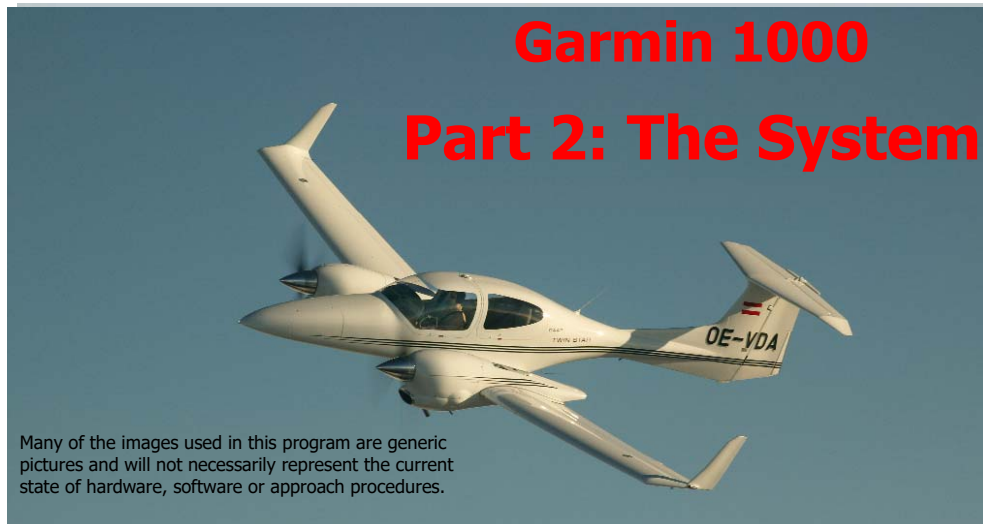




Handout Garmin G1000 Part 2 The System

INTENTIONALLY BLANK

Diamond DA42 Twin Star



Vers. 11.2



DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Content



- [System architecture](#)
- [PFD elements](#)
- [Controls on PFD +MFD](#)
- [MFD elements](#)
- [COM tuning](#)
- [Audio panel](#)
- [NAV tuning](#)
- [CDI](#)
- [Emergency procedures](#)
- [Terrain awareness](#)
- [Traffic avoidance: TIS](#)
- [Traffic avoidance: TAS](#)
- [Stormscope](#)
- [GFC700](#)
- [Safe Taxi, Chart View](#)
- [Synthetic Vision](#)
- [Databases](#)



DIAMOND FLIGHT TRAINING

© Diamond Flight Training

2
Compiled by Peter Schmidleitner

G1000

- The Garmin G1000 is an integrated instrument, communication and navigation system



Garmin 1000



PFD

Primary Flight Display

Audio Panel

MFD

Multi Function Display



Garmin 1000



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

5

Compiled by Peter Schmidleitner

■ General improvements

- Modernised graphics
- Processor speed enhanced
- Database management improved
- Wireless database update with Flight Stream and Garmin Pilot
- Softkey labels and menu-structures rearranged

© Diamond Flight Training

6

Compiled by Peter Schmidleitner

- New Failure indication philosophy
 - Red **X**:
 - Immediate pilot action required
 - Yellow **X**:
 - All other failure indications



■ Color philosophy

CYAN

- Pilot-editable items

WHITE

- Selectable items



G1000 Features



- 10" TFT LCD Displays for the PFD and MFD (1024 x 768 pixels)
- Solid State Attitude-Heading-Reference-System (AHRS)
- Digital Air Data Computer
- Digital Audio Panel
- Dual VHF Communication Radios
- Dual GPS Receivers
- Dual VOR, LOC, GS
- Terrain Awareness
- TIS Traffic Avoidance
- Extensive 3rd party interface capability
 - KAP 140 Autopilot
 - ADF
 - DME
 - TAS
 - Stormscope
 -



G1000 Specifications



- VHF COM Radio - 118.00 MHz to 136.975 MHz with 25kHz or 8.33kHz spacing
- GPS approach capable
- ILS Receiver for precision approaches (Localizer and Glideslope)
- VOR Receiver (108.0 MHz to 117.975 MHz)
- 256 Color, TFT moving map display



Power-up and Self Test



- Aircraft power-up sequence
 - Displays will be on during engine start
 - Red X on various items, e.g. until AHRS aligned, and air data available
 - MFD - verify Jeppesen database dates



Intentionally blank



Garmin G1000



System Architecture

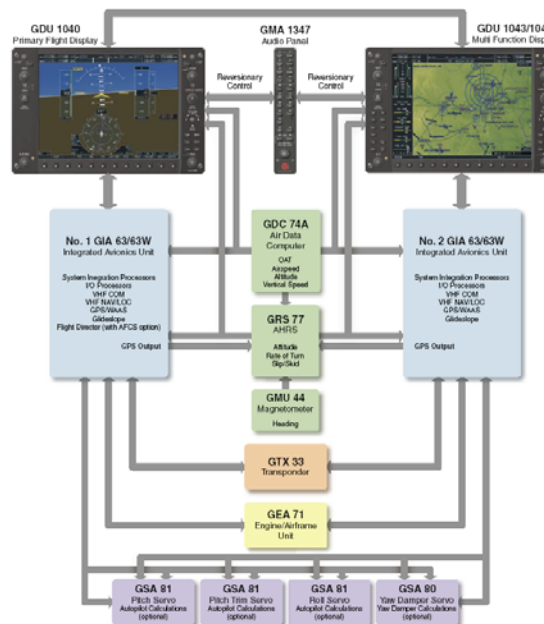


DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

System Architecture



DIAMOND FLIGHT TRAINING

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved

© Diamond Flight Training

14

Compiled by Peter Schmidleitner

System Architecture

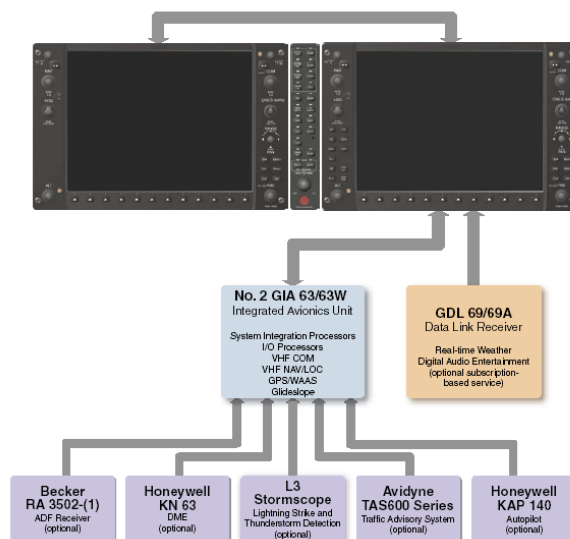


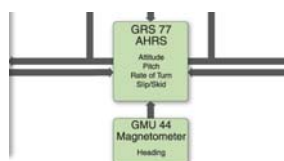
Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved

© Diamond Flight Training

15

Compiled by Peter Schmidleitner

Heading Information



■ Attention !

- Magnetometer output routed through AHRS
- Therefore
- AHRS fail = HDG fail

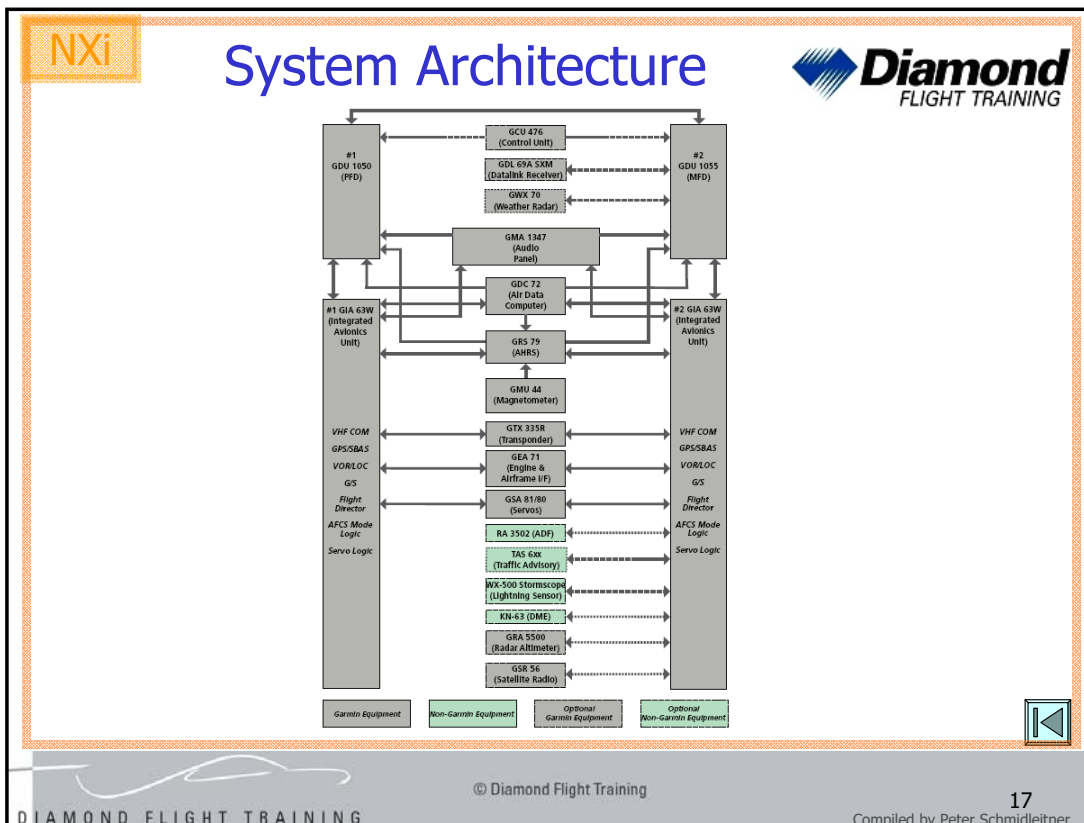


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved

© Diamond Flight Training

16

Compiled by Peter Schmidleitner



G1000/KAP140 Autopilot Interface

Diamond
FLIGHT TRAINING

■ Attention !

- G1000 Altitude Preselect and Baro-setting does not work with the **KAP140** Autopilot
- Use **KAP140** Altitude Preselect control and Baro-setting

© Diamond Flight Training

18
Compiled by Peter Schmidleitner

GIA 63



Communications hub for the system

Primary communications path for AHRS and Air Data information
Interfaces with GDL69 (Sattelite Data Link), Stormscope™,
SkyWatch™, etc.



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

19

Compiled by Peter Schmidleitner

GDC 74 Air Data Computer



Range

Altitude -1,400 ft to 55,000 ft

IAS up to 430 kts

Operate -55°C to +70°C

Inputs include pitot, static and OAT

Accuracy supports RVSM



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

20

Compiled by Peter Schmidleitner

Rapid alignment during taxi

Typically aligns in less than 1 min.!

In-Air Dynamic Restarts

Bank angles up to 20 degrees!

Robust

Operates well in absence of GPS, Air Data, or Magnetometer!

Inherent redundancy / integrity

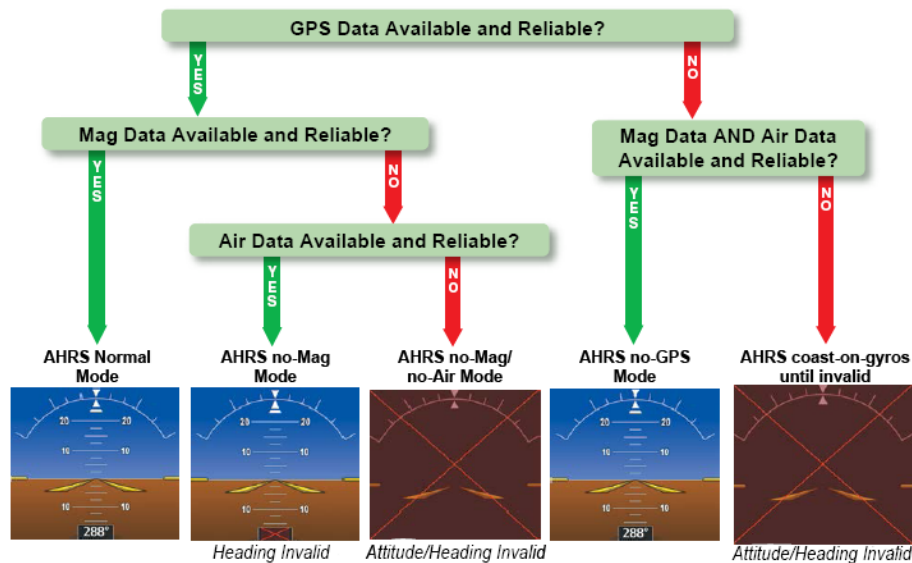


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

21

Compiled by Peter Schmidleitner

AHRS Operation



© Diamond Flight Training

22

Compiled by Peter Schmidleitner

GTX 33 Transponder



- Meets European Level 2 Mode S Criteria
- Solid State - no warm-up time and long life
- Auto-switching between ALT Mode and GND Mode (switched with GPS ground speed input)



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

23

Compiled by Peter Schmidleitner

GEA 71 Engine/Airframe Interface



Extensive input capabilities

Engine load, RPM
Oil temp/pressure
Fuel flow/temp/quantity
Electrical system data
Coolant temp/Gearbox temp

Communicates with both GIA 63s

Brings in not only engine data, but airframe data as well (e.g. Canopy Closed annunciation's)



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

24

Compiled by Peter Schmidleitner

GMA 1347 Digital Audio Panel



Digital audio is used for the Garmin NAVs and COMs

Emergency analog operation that bypasses the GMA 1347

Voice Synthesis / Call-outs - eliminates need for additional hardware

Reversionary Mode button



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

25

Compiled by Peter Schmidleitner



Intentionally blank



© Diamond Flight Training

DIAMOND FLIGHT TRAINING

26

Compiled by Peter Schmidleitner

Garmin G1000



PFD Elements

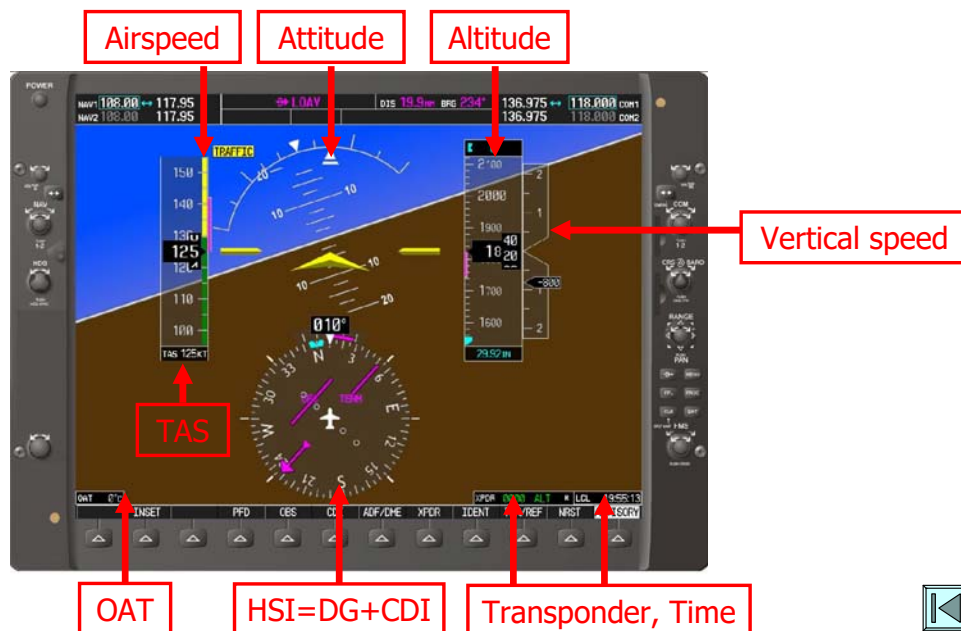


DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

PFD



DIAMOND FLIGHT TRAINING

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved

© Diamond Flight Training

28

Compiled by Peter Schmidleitner

PFD

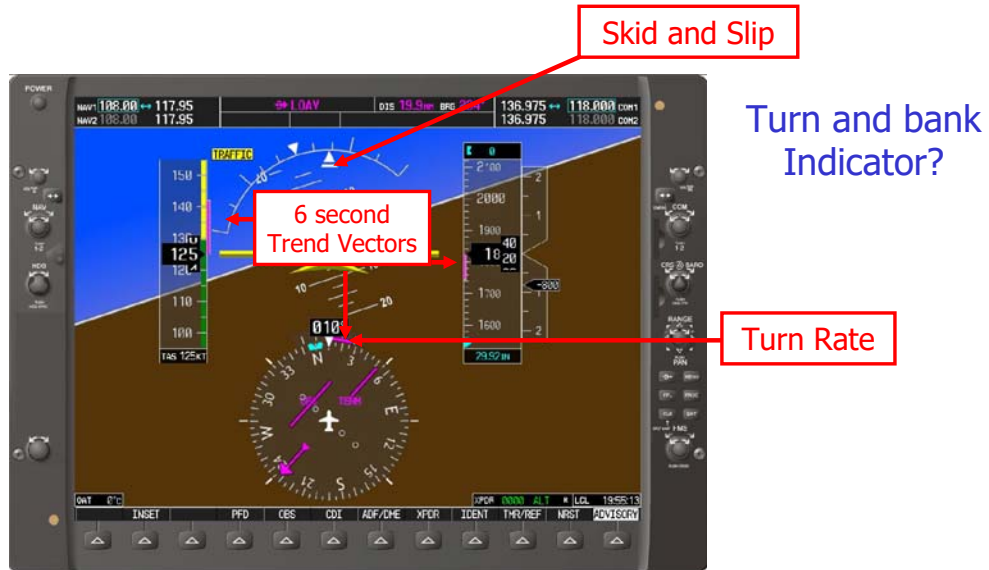


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

29

Compiled by Peter Schmidleitner

PFD



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

30

Compiled by Peter Schmidleitner



Enhanced Map on the HSI

- Pilot selectable, but cannot be used at the same time as an inset map
- zoomable via joystick
- available overlays:
 - Safe Taxi
 - Flight plan
 - Weather Radar
 - Relative Terrain
 - Traffic
 - Obstacles



e.g. Flight Plan:



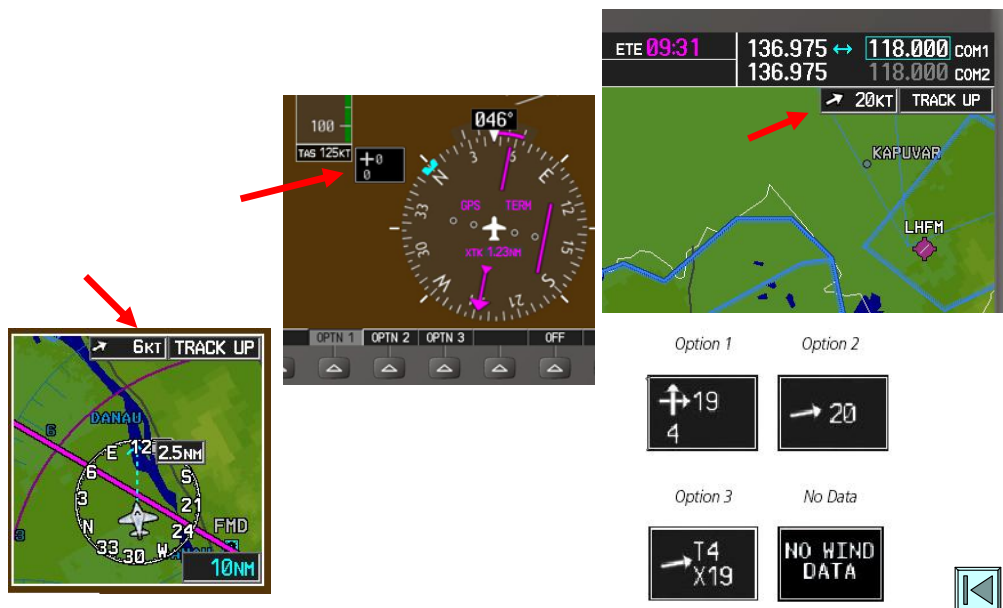
e.g. Relative Terrain:



e.g. Weather Radar:



Wind data display



PFD

PFD NAV status box



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

37

Compiled by Peter Schmidleitner

PFD NAV status box

MCI → TIFTO DIS 51.9 NM BRG 287°

Symbol	Description
→	Active Leg
→	Direct-to
↻	Right Procedure Turn
↻	Left Procedure Turn
↻	Right Holding Pattern
↻	Left Holding Pattern
vtf	Vector to Final
↻	Right DME Arc
↻	Left DME Arc

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

38

Compiled by Peter Schmidleitner

Altitude alert



ALT selected

within 1000 ft

within 200 ft

**Deviating
> 200 ft**

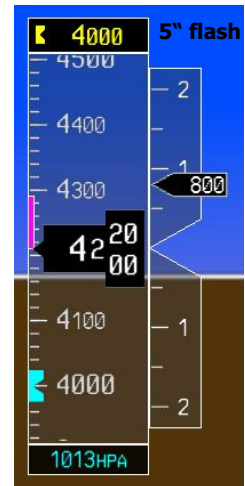
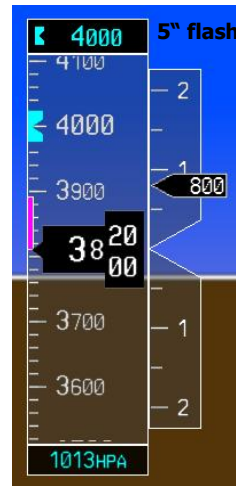
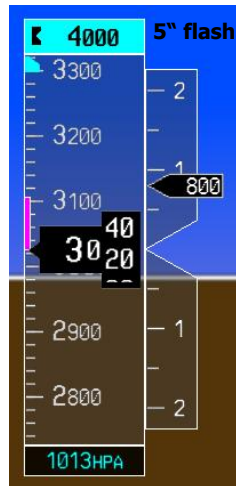
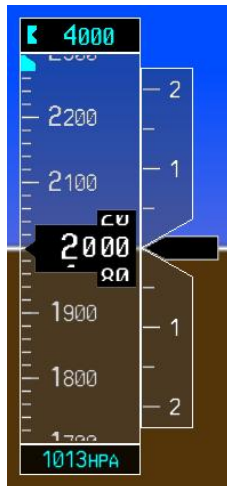


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

39

Compiled by Peter Schmidleitner

DIAMOND FLIGHT TRAINING



Intentionally blank



© Diamond Flight Training

40

Compiled by Peter Schmidleitner

DIAMOND FLIGHT TRAINING

Garmin G1000



PFD Controls



NAV tuning

COM tuning



PFD Controls



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

43

Compiled by Peter Schmidleitner

PFD Controls

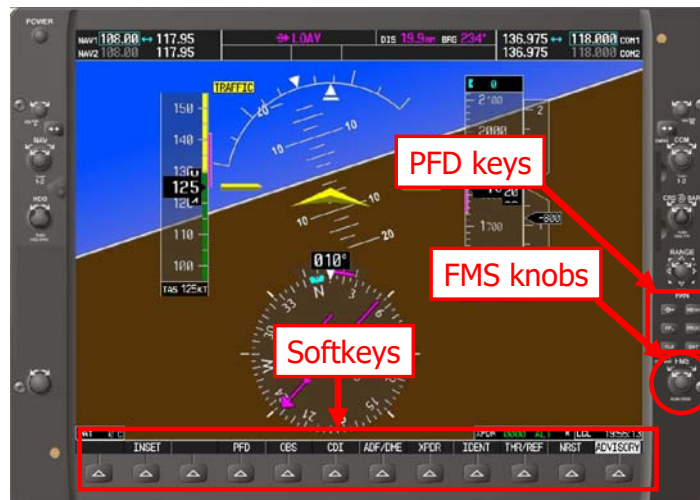


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

44

Compiled by Peter Schmidleitner

MFD Controls

NAV tuning

COM tuning

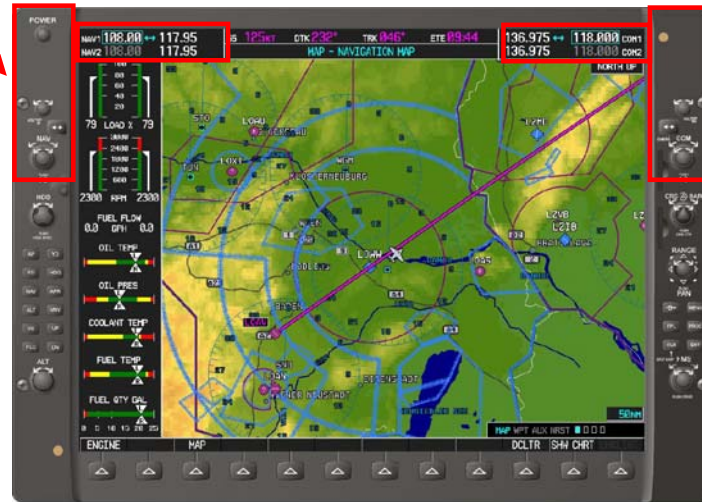


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

45

Compiled by Peter Schmidleitner

MFD Controls

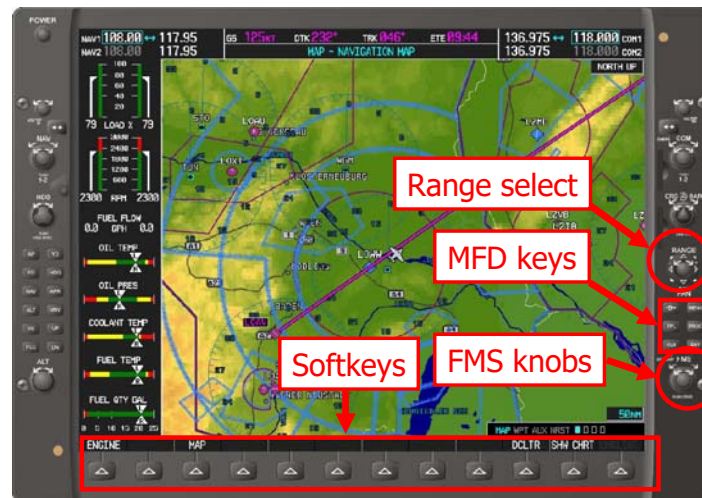


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

46

Compiled by Peter Schmidleitner

MFD Controls

- FMS knobs on MFD have dual function:
 - MFD page select
 - Cursor movement and data input
- To toggle between the functions:
 - Press the small FMS knob

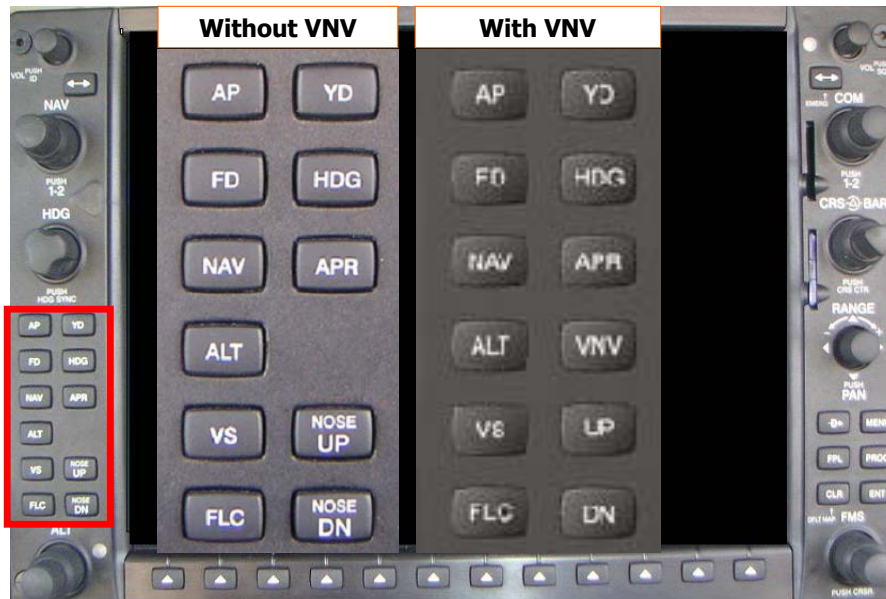


MFD Controls

- FMS knob(s)
 - When cursor is present
 - Large knob changes field
 - Small knob makes input
 - When cursor is **not** present
 - Large knob selects page group
 - Small knob selects page
- Therefore:
 - When you want to change the page:
 - First check that the cursor is removed (press small knob)



MFD Controls with GFC 700



PFD and MFD Controls



This is your working area

Garmin G1000



DIAMOND FLIGHT TRAINING

© Diamond Flight Training

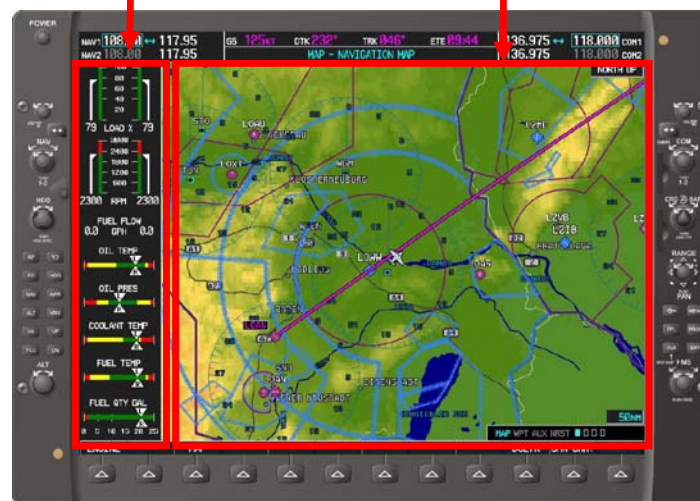
Compiled by Peter Schmidleitner

MFD



EIS / Engine Indication System

Multi Function Display



DIAMOND FLIGHT TRAINING

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

52

Compiled by Peter Schmidleitner

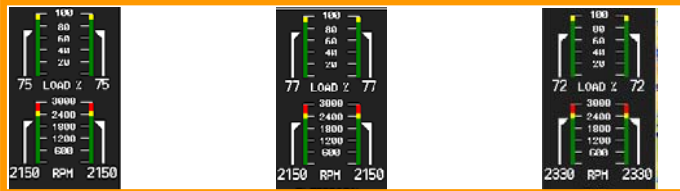
Engine Indication System



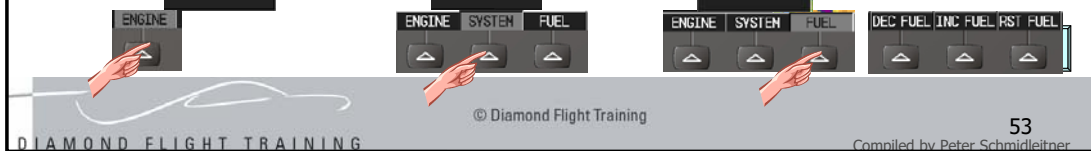
DEFAULT page

SYSTEM page

FUEL page



Upper part always the same



© Diamond Flight Training

53

Compiled by Peter Schmidleitner

Engine Indication System



Default

SYSTEM

FUEL

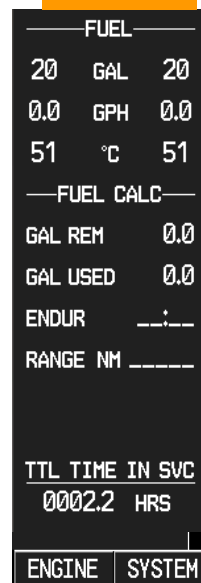
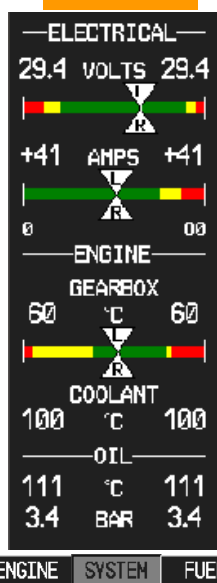
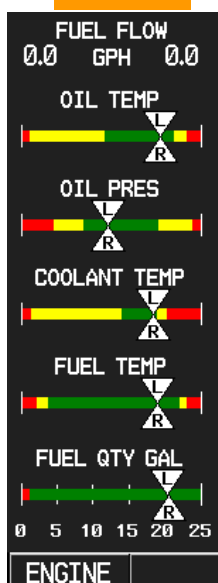


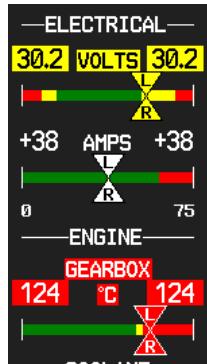
Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved

© Diamond Flight Training

54

Compiled by Peter Schmidleitner

Engine Indication System



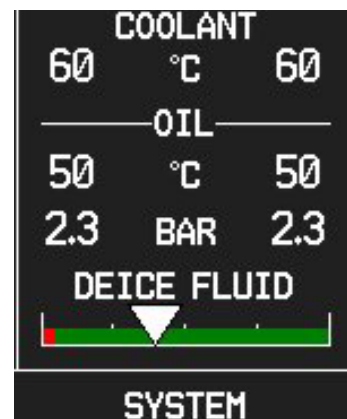
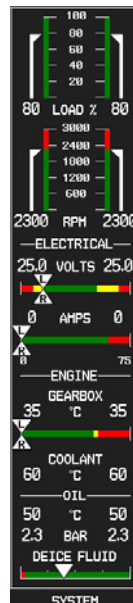
Readouts and labels change colour corresponding to the level of the condition.

Pointers on the horizontal bar indicators:

- white to indicate normal operation
- yellow or red to indicate caution or warning conditions



Ice protection system



NXi

Engine Indication System



© Diamond Flight Training

57

Compiled by Peter Schmidleitner

MFD



NAV status box

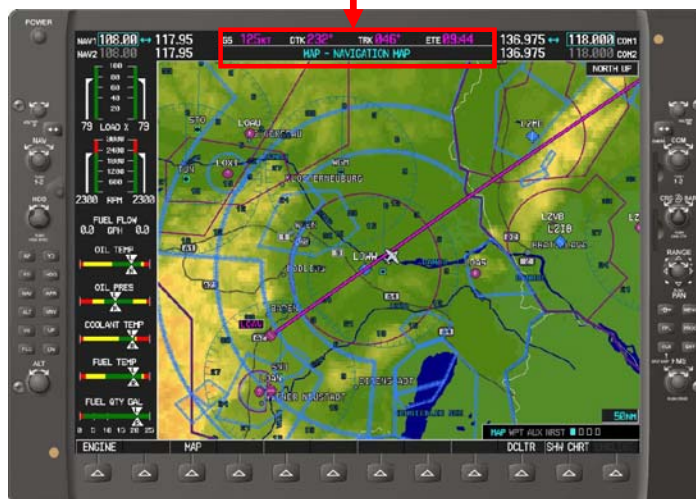


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved

© Diamond Flight Training

58

Compiled by Peter Schmidleitner

MFD NAV status box



GS 141KT DTK 286° TRK 286° ETE 11:47

- Bearing (BRG)
- Distance (DIS)
- Desired Track (DTK)
- Endurance (END)
- Enroute Safe Altitude (ESA)
- Estimated Time of Arrival (ETA)
- Estimated Time Enroute (ETE)
- Fuel on Board (FOB)
- Fuel over Destination (FOD)
- Ground Speed (GS)
- Minimum Safe Altitude (MSA)
- True Air Speed (TAS)
- Track Angle Error (TKE)
- Track (TRK)
- Vertical Speed Required (VSR)
- Crosstrack Error (XTK)



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

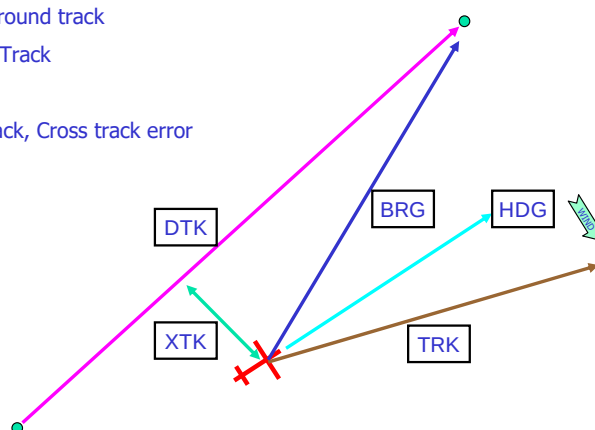
59

Compiled by Peter Schmidleitner

Are you on the right track?



- HDG Heading
TRK Actual ground track
DTK Desired Track
BRG Bearing
XTK Cross track, Cross track error



© Diamond Flight Training

DIAMOND FLIGHT TRAINING

60

Compiled by Peter Schmidleitner

MFD

NAV frequencies

NAV Status Box

COM frequencies



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

61

Compiled by Peter Schmidleitner

MFD

The Page Group Icon shows which page group and page you are viewing.

Page group indicator

Selected page indicator

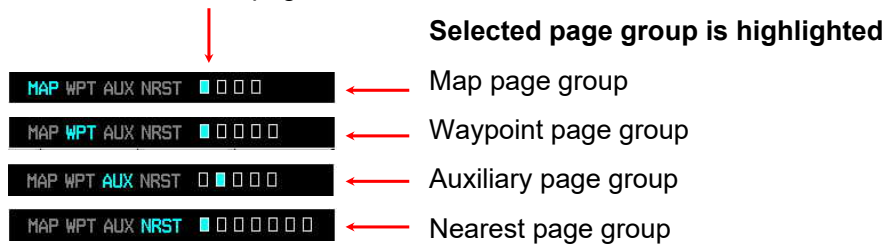


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

62

Compiled by Peter Schmidleitner

NXi

MFD





Map - Navigation Map

Page Group and Page Name

Pages in the Group

Page Group

Navigation Map
 Traffic Map
 Weather Radar
 Stormscope®
 Weather Data Link
 Terrain-SVT
Map WPT Aux FPL NRST

DIAMOND FLIGHT TRAINING

© Diamond Flight Training

63
Compiled by Peter Schmidleitner

MFD

Map Page Group





1. Navigation Map



2. Traffic Map



3. Weather Map



4. Terrain Proximity

DIAMOND FLIGHT TRAINING

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
 © Diamond Flight Training

64
Compiled by Peter Schmidleitner

MFD

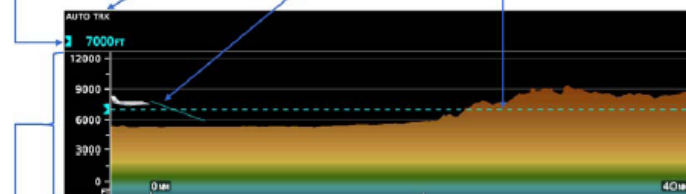
Navigation Map Profile View



NXi

■ Vertical Situation Display instead of Vertical Profile View

Selected altitude readout Mode indication Vertical track vector Selected altitude line



Altitude range changes with map range

- Limited to keep altitude range usable at high map ranges



Vertical Situation Display

■ Available Modes:

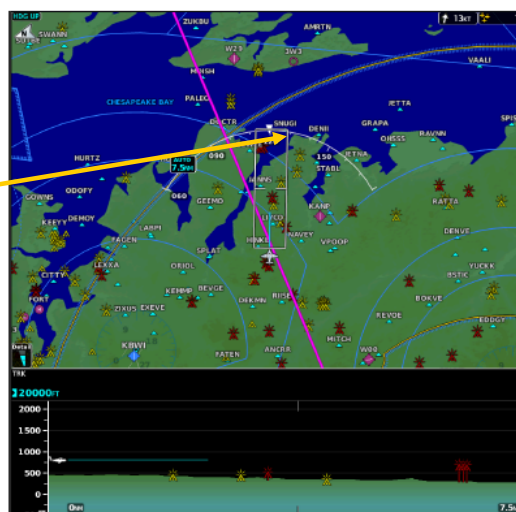
- Track
 - Swath *) centered and aligned with current track
- Flight Plan:
 - Swath centered and aligned with flight plan
- Auto:
 - VSD selects Flight Plan when within flight plan swath and Track when outside flight plan swath

*) Swath: the Vertical Situation Display path.

Here in track mode:



- Track Mode Boundary on Navigation Map
- Vertical Situation Display Track Mode



Waypoint labels

Altitude constraint values

VNAV descent path

Altitude constraint symbols

Airspace Altitude Labels:

Upper and lower airspace boundaries on the depicted airspace



MAP orientation logic



- Map orientation is selected on Navigation Map page
- Maps showing navigation data are in the orientation selected for the Navigation Map page
- Other maps have their own, logical map orientation, e.g. HDG UP or NORTH UP



NXi

Map Setup



Automatic switching to NORTH UP



- Weather Datalink page
map orientation synchronised
with NAV map orientation



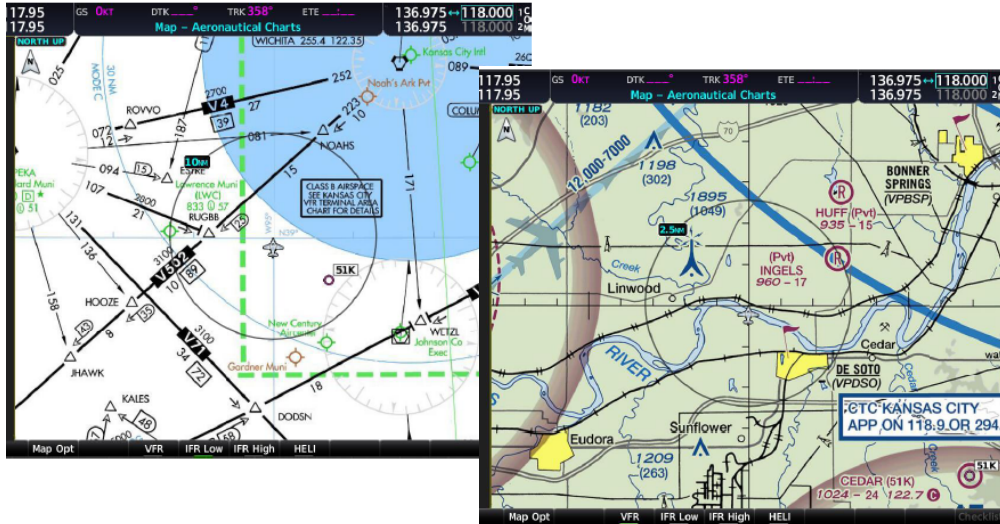
Selected Altitude Intercept Arc

MAP SETUP	
GROUP	
Map	
ORIENTATION	North up
AUTO ZOOM	OFF
MAX LOOK FWD	30min
MIN LOOK FWD	5min
TIME OUT	0min
LAND DATA	< On >
TRACK VECTOR	< OFF > 60 sec
WIND VECTOR	< OFF >
NAV RANGE RING	< OFF >
TOPO DATA	< On > 1500NM
TOPO SCALE	< On >
TERRAIN DATA	< OFF > 2000NM
OBSTACLE DATA	< OFF > 20NM
FUEL RING (RSV)	< OFF > 00:45
FIELD OF VIEW	< OFF >
SEL ALT ARC	< OFF >

Range to
Altitude Arc



■ IFR / VFR Enroute Charts



© Diamond Flight Training

75

Compiled by Peter Schmidleitner

Decluttering the map

Declutter-1

removed:

Land/Country Text

Cities

Roads

Railroads

State/Province Boundaries

River/Lake Names

Item	No Declutter	Declutter-1	Declutter-2	Declutter-3
Flight Plan Route Lines	X	X	X	X
Flight Plan Route Waypoints	X	X	X	X
Rivers/Lakes	X	X	X	X
Topography Data	X	X	X	X
International Borders	X	X	X	X
Land/Country Text	X	X	X	X
Cities	X	X	X	X
Roads	X	X	X	X
Railroads	X	X	X	X
State/Province Boundaries	X	X	X	X
River/Lake Names	X	X	X	X
User Waypoints	X	X		
Latitude/Longitude Grid	X	X		
NAVAIDS	X	X		
Class B Airspaces/TMA	X	X		
Class C Airspaces/TCA	X	X		
Class D Airspaces	X	X		
Other Airspaces/ADIZ	X	X		
TFRs	X	X		
Obstacles	X	X		
Land/Country Text	X			
Cities	X			
Roads	X			
Railroads	X			
State/Province Boundaries	X			
River/Lake Names	X			



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved

© Diamond Flight Training

76

Compiled by Peter Schmidleitner

Decluttering the map

Item	No Declutter	Declutter-1	Declutter-2	Declutter-3
Flight Plan Route Lines	X	X	X	X
Flight Plan Route Waypoints	X	X	X	X
Rivers/Lakes	X	X	X	X
Topography Data	X	X	X	X
International Borders	X	X	X	X
Track Vector	X	X	X	X
Navigation Range Ring	X	X	X	X
User Waypoints	X	X	X	X
Latitude/Longitude Grid	X	X	X	X
NAVAIDs	X	X	X	X
Class B Airspaces/TMA	X	X	X	X
Class C Airspaces/TCA	X	X	X	X
Class D Airspaces	X	X	X	X
Other Airspaces/ADIZ	X	X	X	X
TFRs	X	X	X	X
Obstacles	X	X	X	X
Obstacles	X	X	X	X
Land/Country Text	X	X	X	X
Cities	X	X	X	X
Roads	X	X	X	X
Railroads	X	X	X	X
State/Province Boundaries	X	X	X	X
River/Lake Names	X	X	X	X



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

77

Compiled by Peter Schmidleitner

Decluttering the map

Item	No Declutter	Declutter-1	Declutter-2	Declutter-3
Flight Plan Route Lines	X	X	X	X
Flight Plan Route Waypoints	X	X	X	X
Rivers/Lakes	X	X	X	X
Topography Data	X	X	X	X
International Borders	X	X	X	X
Track Vector	X	X	X	X
Navigation Range Ring	X	X	X	X
NEXRAD	X	X	X	X
XM Lightning Data	X	X	X	X
Airports	X	X	X	X
Runway Labels	X	X	X	X
Restricted	X	X	X	X
MOA (Military)	X	X	X	X
Obstacles	X	X	X	X
Land/Country Text	X	X	X	X
Cities	X	X	X	X
Roads	X	X	X	X
Railroads	X	X	X	X
State/Province Boundaries	X	X	X	X
River/Lake Names	X	X	X	X



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

78

Compiled by Peter Schmidleitner

Decluttering the map

always available:

Item	No Declutter	Declutter-1	Declutter-2	Declutter-3
Flight Plan Route Lines	X	X	X	X
Flight Plan Route Waypoints	X	X	X	X
Rivers/Lakes	X	X	X	X
Topography Data	X	X	X	X
International Borders	X	X	X	X
Track Vector	X	X	X	X
Navigation Range Ring	X	X	X	X
Fuel Range Ring	X	X	X	X
Terrain Data	X	X	X	X
Traffic	X	X	X	X
Airways	X	X	X	X
Lanes	X	X	X	X
Roads	X	X	X	X
Railroads	X	X	X	X
State/Province Boundaries	X	X	X	X
River/Lake Names	X	X	X	X



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

79

Compiled by Peter Schmidleitner

MFD Waypoint (WPT) Page Group



1. Airport Information



2. Intersection Information



3. NDB Information



4. VOR Information



5. User WPT Information

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

80

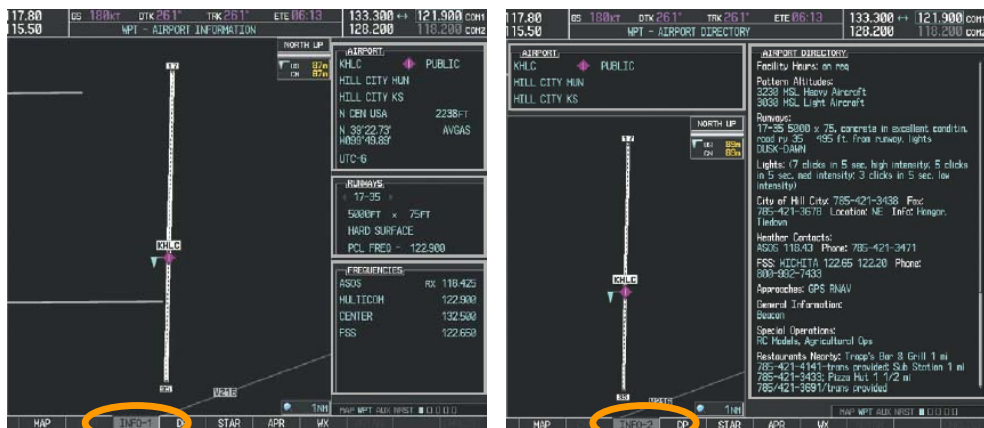
Compiled by Peter Schmidleitner

MFD

Waypoint Page Group



•WPT – Airport Information



AOPA Airport Directory (if available)



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

81

Compiled by Peter Schmidleitner

MFD

Aux Page Group



1. Trip Planning



2. Utility



3. GPS Status



4. System Setup



5. System Status



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

82

Compiled by Peter Schmidleitner

MFD

Nearest Page Group

Nearest...

- Airports
- Intersections
- NDBs
- VORs
- User Waypoints
- ARTCC/FSS/WX
- Airspaces

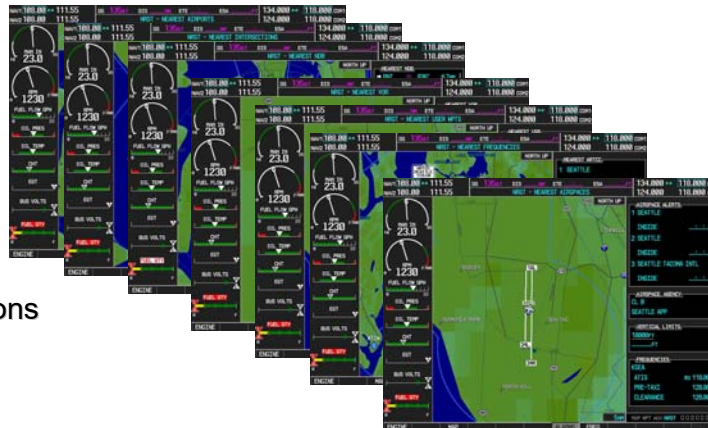


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

85

Compiled by Peter Schmidleitner

DIAMOND FLIGHT TRAINING

MFD

Flight Plan Page Group



1. Active Flight Plan



2. Flight Plan Catalog



© Diamond Flight Training

86

Compiled by Peter Schmidleitner

DIAMOND FLIGHT TRAINING

Flight plan display

Narrow view, leg-leg

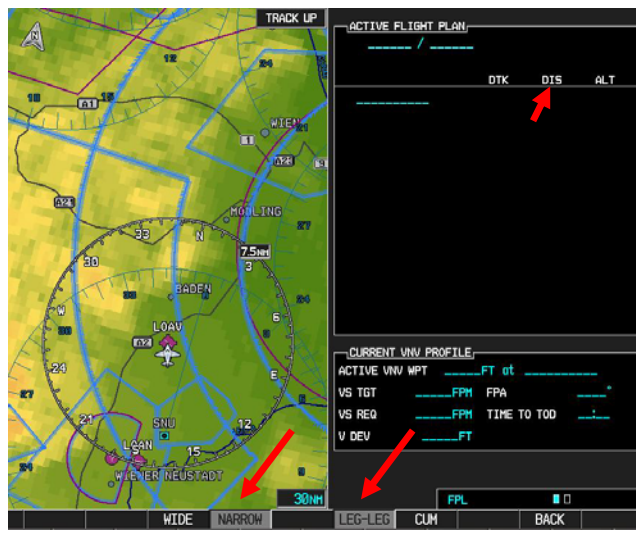


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

87

Compiled by Peter Schmidleitner

Flight plan display

Wide view, leg-leg



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

88

Compiled by Peter Schmidleitner

Flight plan display

Narrow view, cumulative

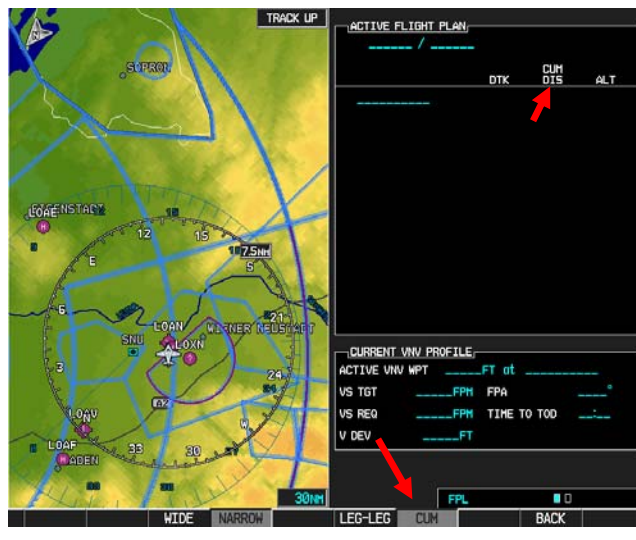


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

89

Compiled by Peter Schmidleitner

Flight plan display

Wide view, cumulative



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

90

Compiled by Peter Schmidleitner

■ Flight Plan Progress Summary

can be added to
the NAV Map
(instead of
Vertical Situation
Display)



- **Runway** may be selected for departure and detination airport
 - If RWY slected for destination, system will navigate to threshold rather than to aerodrome reference point

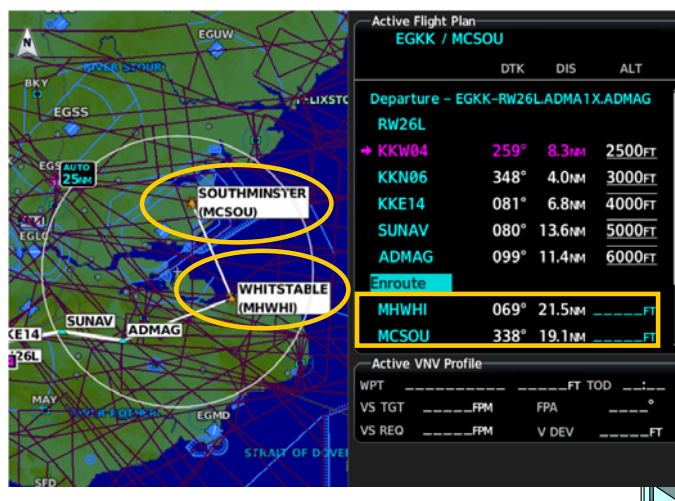


Runway selection
pop up

Runway display
in Flight Plan

Visual Reporting Points

Symbol:



Garmin G1000



NAV and COM Tuning



LH

identical

RH



NAV and COM Tuning

NAV

COM



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

97

Compiled by Peter Schmidleitner

COM Tuning

Active: inside

Standby: outside

Active: green

Standby cursor: boxed

COM
volume/squelch

COM



Active („green“) is selected
on AUDIO panel

COM flip-flop

Cursor toggle

MHz tuning

KHz tuning

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

98

Compiled by Peter Schmidleitner

COM Volume

COM Volume knob operation shows the % of volume

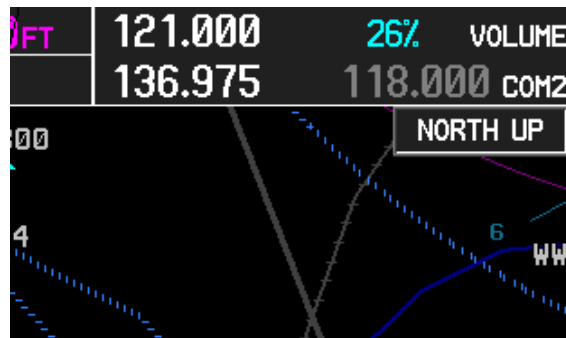


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

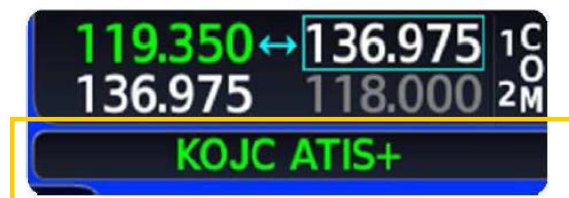
DIAMOND FLIGHT TRAINING

99

Compiled by Peter Schmidleitner

NXi

- **COM frequency decoding**
 - Name of the ATC agency indicated



© Diamond Flight Training

DIAMOND FLIGHT TRAINING

100

Compiled by Peter Schmidleitner

Garmin G1000



Audio Panel



DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Audio Panel



„MIC“ selector; automatically selects corresponding Audio to headphone

COM Split Switch

GMA 1347

Intercom Isolation



DIAMOND FLIGHT TRAINING

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

102

Compiled by Peter Schmidleitner

Audio Panel



Clearance recorder playback

- Last 2,5 minutes of communication
- Press MKR/Mute to exit

„MAN Squelch“ key

Intercom volume and
squelch control



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

103

Compiled by Peter Schmidleitner

Audio Panel

Normal Position: Intercom volume control



MAN SQ off
(not lit)

Auto squelch is
active

Inner knob:
Pilot's volume

Outer knob:
Copilot's, Passengers'
volume



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

104

Compiled by Peter Schmidleitner

Audio Panel

Manual squelch control

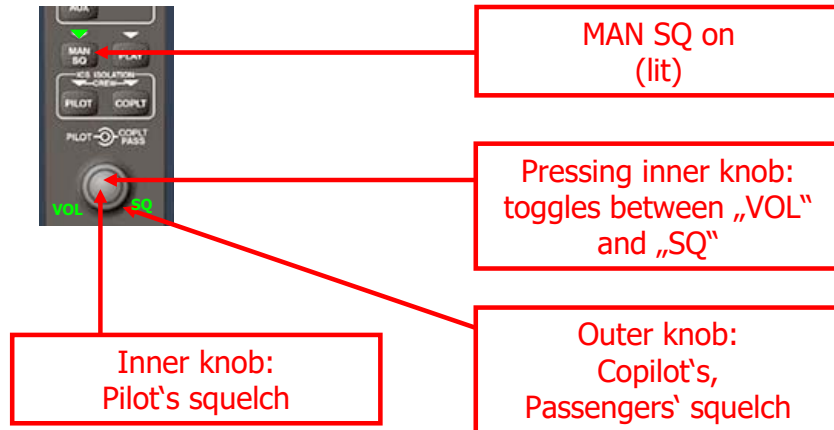


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

105

Compiled by Peter Schmidleitner

Audio Panel

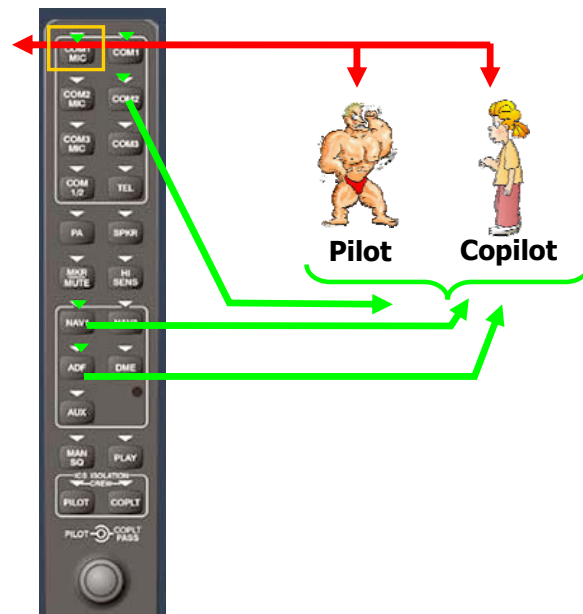


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

106

Compiled by Peter Schmidleitner

Audio Panel

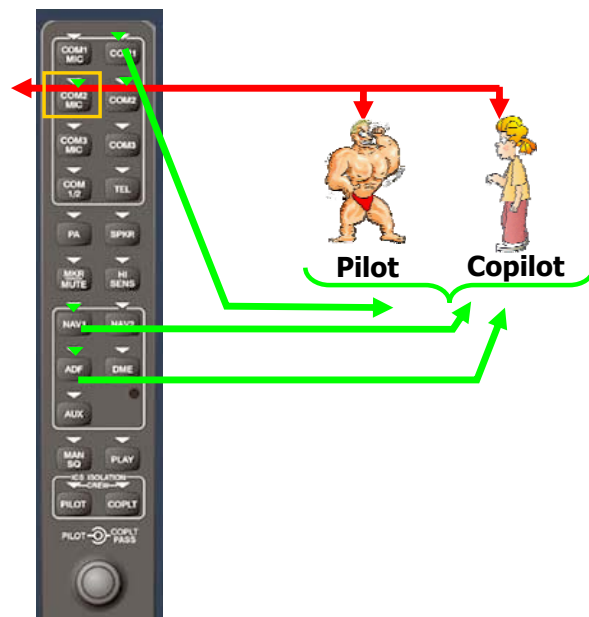


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

107

Compiled by Peter Schmidleitner

Audio Panel

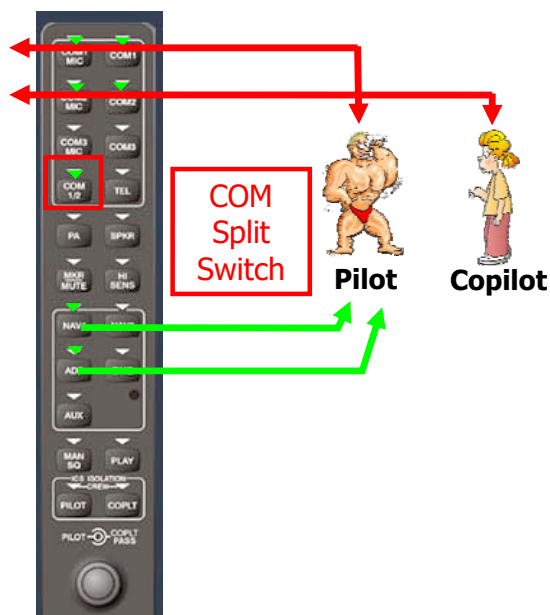


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

108

Compiled by Peter Schmidleitner

Audio Panel

Intercom Isolation



„ALL“
„PILOT“
„COPILOT“
„CREW“



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

109

Compiled by Peter Schmidleitner

Audio Panel

Intercom Isolation

„ALL“

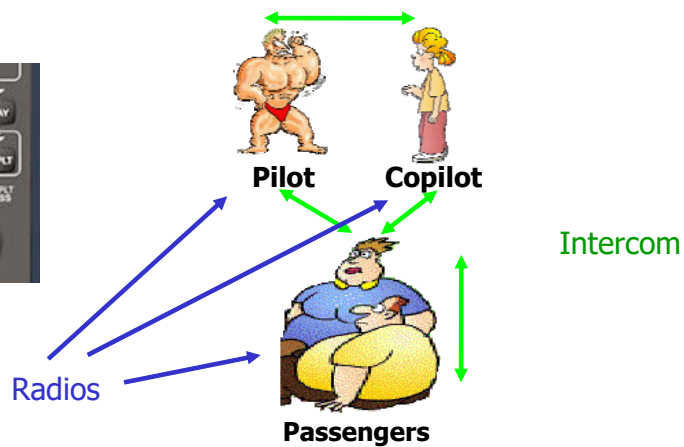


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

110

Compiled by Peter Schmidleitner

Audio Panel

Intercom Isolation

„PILOT“



Radios



Pilot



Copilot



Passengers

Intercom



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

111

Compiled by Peter Schmidleitner

Audio Panel

Intercom Isolation

„COPILOT“



Radios



Pilot



Copilot



Passengers

Intercom



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

112

Compiled by Peter Schmidleitner

Audio Panel

Intercom Isolation

„CREW“

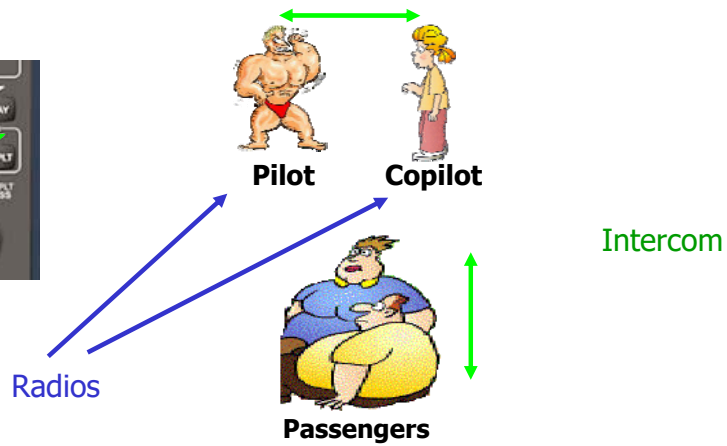


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

113

Compiled by Peter Schmidleitner

DIAMOND FLIGHT TRAINING

Intentionally blank



© Diamond Flight Training

114

Compiled by Peter Schmidleitner

DIAMOND FLIGHT TRAINING

Garmin G1000



NAV Tuning ADF/DME Tuning RMI



DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

NAV Tuning



NAV volume
ident filter

NAV

NAV flip-flop

Cursor toggle

MHz tuning

KHz tuning

Standby: outside

Active: inside

Standby cursor: boxed

Active: green

NAV1 109.90 ↔ 114.10 BNA

NAV2 110.60 111.60 GHM

Active („green”) is on HSI,
selected by CDI key

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved

© Diamond Flight Training

116

Compiled by Peter Schmidleitner

ADF - DME Tuning

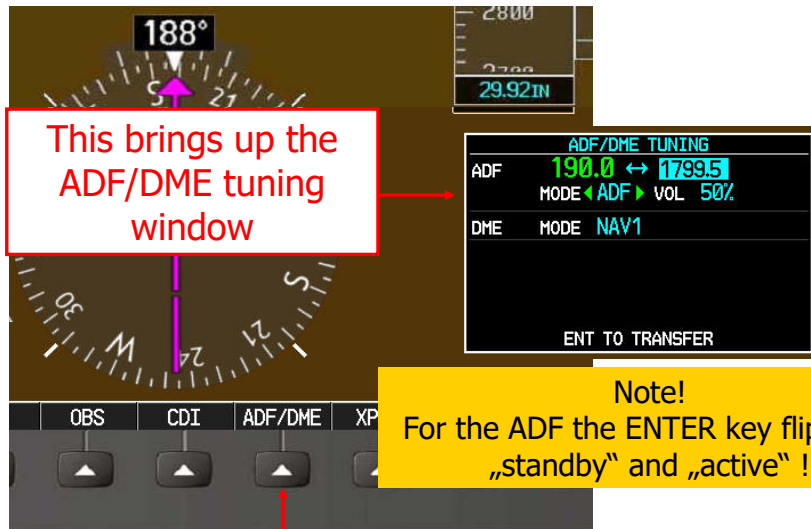


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

117

Compiled by Peter Schmidleitner

DME



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

118

Compiled by Peter Schmidleitner

RMI Needles

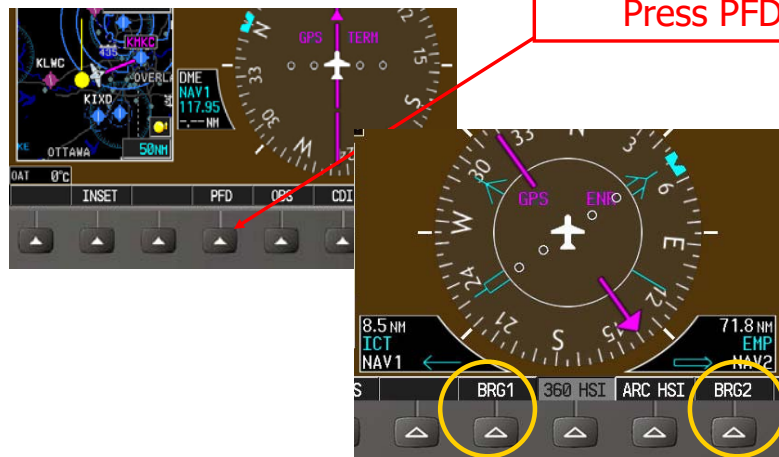


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

119

Compiled by Peter Schmidleitner

RMI Needles



„BRG1“ and „BRG2“
toggle the arrows
between GPS, NAV and
ADF



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

120

Compiled by Peter Schmidleitner

Garmin G1000



CDI Softkey



CDI softkey toggles
between

GPS (magenta)

NAV1 (single green)

NAV2 (double green)



CDI Softkey



CDI softkey toggles
between
GPS (magenta)
NAV1(single green)
NAV2 (double green)



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

123

Compiled by Peter Schmidleitner

CDI Softkey



CDI softkey toggles
between
GPS (magenta)
NAV1(single green)
NAV2 (double green)



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

124

Compiled by Peter Schmidleitner

Why this CDI switching?



Classic installation



NAV #1

NAV #2

DME

ADF



CDI Softkey

G1000 installation



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

127

Compiled by Peter Schmidleitner

Track indicator

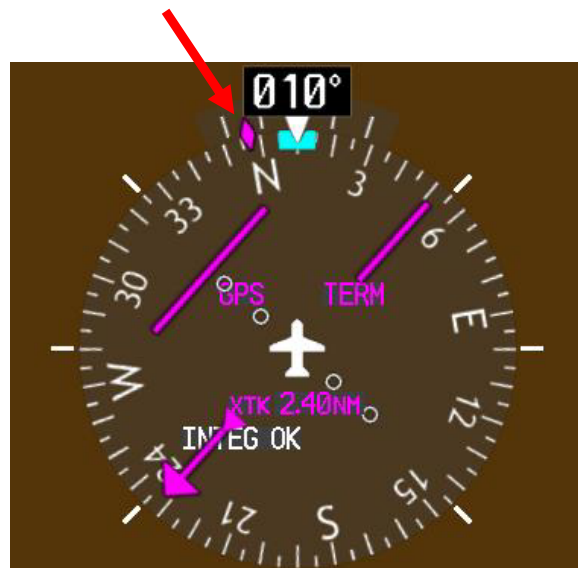


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

128

Compiled by Peter Schmidleitner

XTK indication



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

129

Compiled by Peter Schmidleitner

Flight phase indication



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

130

Compiled by Peter Schmidleitner

Flight phase indication

Flight Phase	Annunciation*	Automatic CDI Full-scale Deflection
Departure	DPRT	0.3 nm
Terminal	TERM	1.0 nm
Enroute	ENR	2.0 nm
Oceanic	OCN	2.0 nm
Approach (Non-precision)	LNAV	1.0 nm decreasing to 350 feet
Approach (Non-precision with Vertical Guidance)	LNAV + V	SBAS necessary
Approach (LNAV/VNAV)	L/VNAV	1.0 nm decreasing to a specified course width
Approach (LPV)	LPV	
Missed Approach	MAPR	0.3 nm



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

131

Compiled by Peter Schmidleitner

DIAMOND FLIGHT TRAINING

Intentionally blank



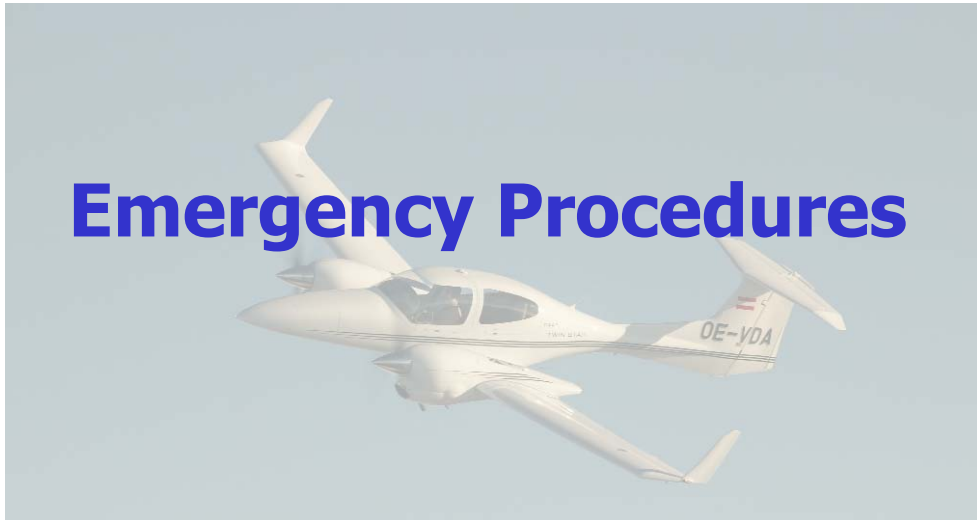
© Diamond Flight Training

132

Compiled by Peter Schmidleitner

DIAMOND FLIGHT TRAINING

Garmin G1000



DIAMOND FLIGHT TRAINING

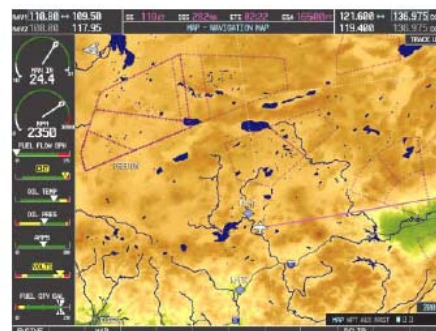
© Diamond Flight Training

Compiled by Peter Schmidleitner

Emergency Procedures



Normal Mode Display



DIAMOND FLIGHT TRAINING

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved

© Diamond Flight Training

134

Compiled by Peter Schmidleitner

Emergency Procedures



PFD Failure



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

135

Compiled by Peter Schmidleitner

Emergency Procedures



MFD Failure



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

136

Compiled by Peter Schmidleitner

Emergency Procedures

Reversionary Mode



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

137

Compiled by Peter Schmidleitner

Emergency Procedures

Alerting System

System Annunciations

Failure indicated with a red cross

Displays abbreviated annunciation text

Displays alert text messages

Annunciation Window

Flashes as „WARNING“, „CAUTION“ or „ADVISORY“

Alerts Window

Alerts Softkey Annunciation

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

138

Compiled by Peter Schmidleitner

Emergency Procedures



Air Data Computer Fail



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

139

Compiled by Peter Schmidleitner

Emergency Procedures



AHRS Fail



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

140

Compiled by Peter Schmidleitner

Emergency Procedures



Magnetometer Fail



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
 © Diamond Flight Training

141

Compiled by Peter Schmidleitner

Emergency Procedures



Failed System	Action
AHRS	Refer to SBY horizon, attempt AHRS reset
ADC	Refer to SBY ASI, Altimeter
NAV / COM	Use other NAV / COM
Audio panel	Automatic backup to analog mode of COM 1, no intercom



© Diamond Flight Training

142

Compiled by Peter Schmidleitner

Integrity annunciations

Annunciation	Description
LOI	Loss of Integrity Monitoring–GPS integrity is insufficient for the current phase of flight
INTEG OK	Integrity OK–GPS integrity has been restored to within normal limits (annunciation displayed for 5 seconds)
DR	Dead Reckoning–System is using projected position rather than GPS position to compute navigation data and sequence active flight plan waypoints

Dead Reckoning available in ENROUTE or OCEANIC phase only

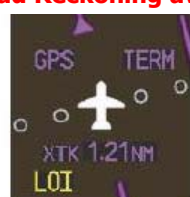


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

143

Compiled by Peter Schmidleitner

DIAMOND FLIGHT TRAINING

System message advisories

LOI – GPS integrity lost. Crosscheck with other NAVS.	GPS integrity is insufficient for the current phase of flight.
GPS NAV LOST – Loss of GPS navigation. Insufficient satellites.	Loss of GPS navigation due to insufficient satellites.
GPS NAV LOST – Loss of GPS navigation. Position error.	Loss of GPS navigation due to position error.
GPS NAV LOST – Loss of GPS navigation. GPS fail.	Loss of GPS navigation due to GPS failure.
ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
APR DWNGRADE – Approach downgraded.	Use LNAV minima when approach is downgraded.



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

144

Compiled by Peter Schmidleitner

DIAMOND FLIGHT TRAINING

Garmin G1000



Terrain Awareness



DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Terrain Awareness



- Terrain Awareness needs
 - a valid 3D GPS position
 - a valid terrain/obstacle database



DIAMOND FLIGHT TRAINING

© Diamond Flight Training

146
Compiled by Peter Schmidleitner

Terrain Awareness

- For situational awareness only
- No voice callout
- Yellow = aircraft within 1000' AGL of terrain
- Red = aircraft below elevation of terrain
- does not account for trees, power-lines, etc.



Terrain Awareness

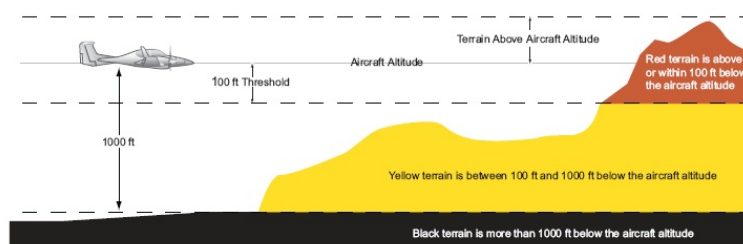


Figure 6-44 Terrain Altitude/Color Correlation for Terrain Proximity

Unlighted Obstacle		Lighted Obstacle		Obstacle Location
< 1000' AGL	> 1000' AGL	< 1000' AGL	> 1000' AGL	
				Red obstacle is above or within 100 ft below the aircraft altitude
				Yellow obstacle is between 100 ft and 1000 ft below the aircraft altitude

Table 6-4 Terrain Proximity Terrain/Obstacle Colors and Symbology



Terrain Awareness

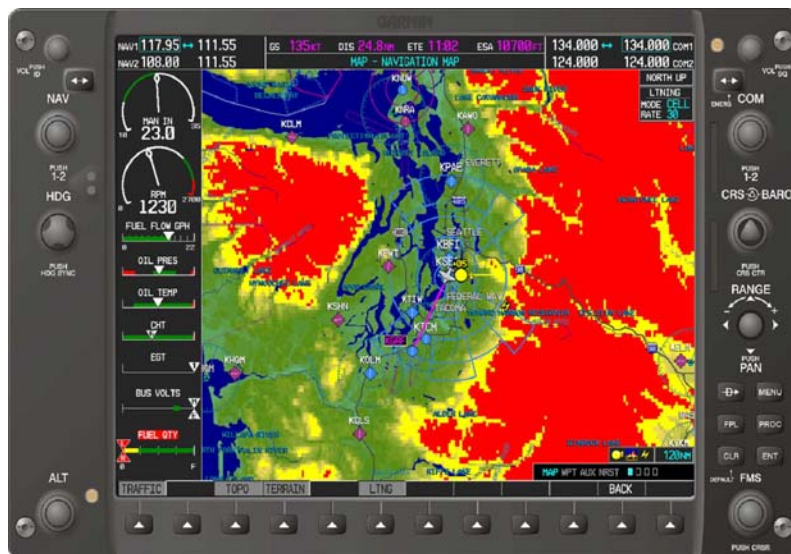


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

149

Compiled by Peter Schmidleitner

DIAMOND FLIGHT TRAINING

Terrain Awareness Profile View

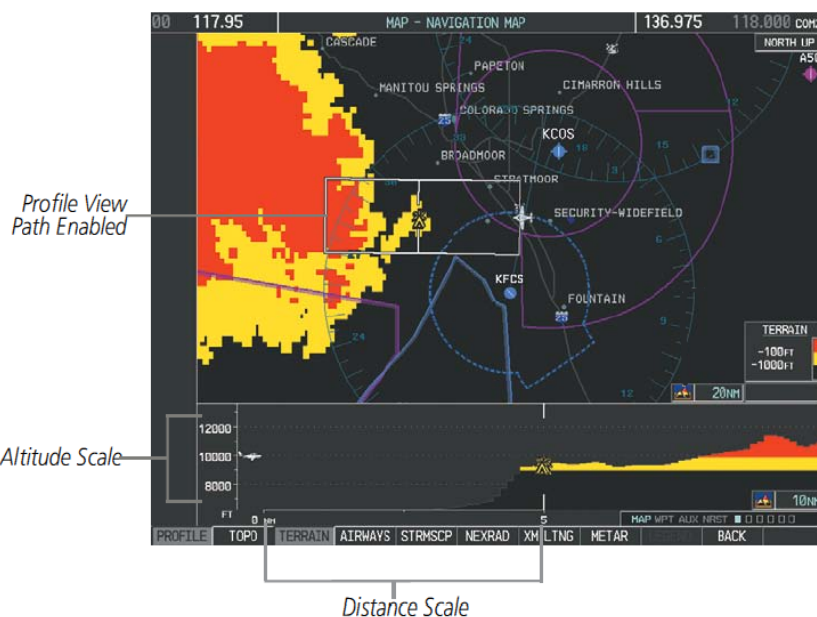


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

150

Compiled by Peter Schmidleitner

DIAMOND FLIGHT TRAINING

Terrain-SVS



Terrain-SVS

- Terrain-SVS is integrated in SVT (Synthetic Vision Technology)
- Terrain-SVS needs
 - a valid 3D GPS position
 - a valid terrain/obstacle database



Terrain-SVS

- It provides visual and auditory alerts to indicate threatening terrain relevant to the projected flight path
 - But it is not a Terrain Awareness and Warning System (TAWS), it does not provide
 - Premature Descent Alerting
 - Excessive Descent Rate
 - Negative Climb Rate
 - Descent to 500 Feet Voice Callout



Terrain-SVS

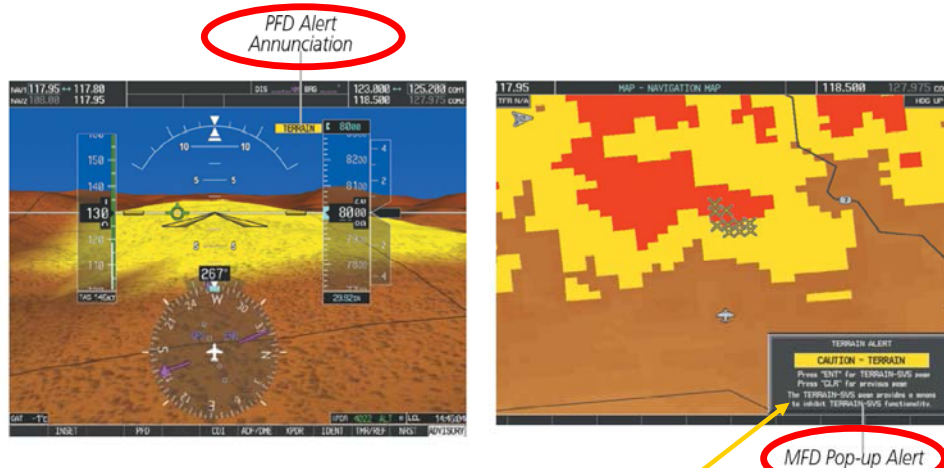


Figure 6-174 Terrain-SVS Alert Annunciations

CLR removes pop-up alert

ENT displays Terrain-SVS page



Terrain Awareness with SVS



Alert Type	PFD/MFD TERRAIN-SVS Page Annunciation	Pop-Up Alert (MFD Navigation Map Page)	Aural Message
Reduced Required Terrain Clearance Warning (RTC)	TERRAIN	WARNING TERRAIN	"Warning; Terrain, Terrain"
Imminent Terrain Impact Warning (ITI)	TERRAIN	WARNING TERRAIN	"Warning; Terrain, Terrain"
Reduced Required Obstacle Clearance Warning (ROC)	TERRAIN	WARNING OBSTACLE	"Warning; Obstacle, Obstacle"
Imminent Obstacle Impact Warning (IOI)	TERRAIN	WARNING OBSTACLE	"Warning; Obstacle, Obstacle"
Reduced Required Terrain Clearance Caution (RTC)	TERRAIN	CAUTION TERRAIN	"Caution; Terrain, Terrain"
Imminent Terrain Impact Caution (ITI)	TERRAIN	CAUTION TERRAIN	"Caution; Terrain, Terrain"
Reduced Required Obstacle Clearance Caution (ROC)	TERRAIN	CAUTION OBSTACLE	"Caution; Obstacle, Obstacle"
Imminent Obstacle Impact Caution (IOI)	TERRAIN	CAUTION OBSTACLE	"Caution; Obstacle, Obstacle"



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

155

Compiled by Peter Schmidleitner

DIAMOND FLIGHT TRAINING

Terrain-SVS



- Forward Looking Terrain Avoidance (FLTA) issues visual and voice alerts when the projected flight path conflicts with terrain or obstacles



© Diamond Flight Training

156

Compiled by Peter Schmidleitner

DIAMOND FLIGHT TRAINING

Terrain Awareness with SVS

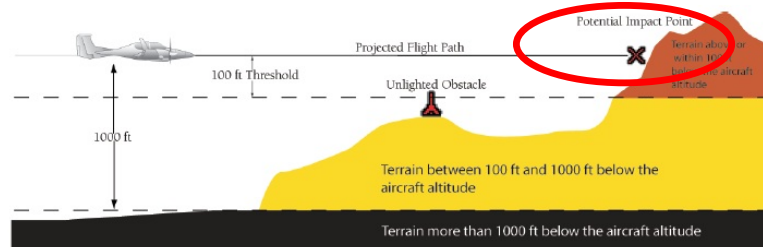


Figure 6-51 Terrain Altitude/Color Correlation for TERRAIN-SVS

Unlighted Obstacle		Lighted Obstacle		Potential Impact Points	Obstacle Location
< 1000' AGL	> 1000' AGL	< 1000' AGL	> 1000' AGL		
					WARNING: Red obstacle is above or within 100' below current aircraft altitude
					CAUTION: Yellow obstacle is between 100' and 1000' below current aircraft altitude

Table 6-5 TERRAIN-SVS Obstacle Colors and Symbology



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

157

Compiled by Peter Schmidleitner

DIAMOND FLIGHT TRAINING

Terrain Awareness with SVS

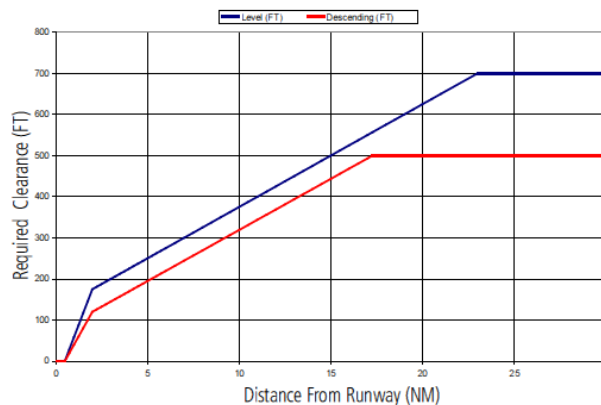


Figure 6-76 FLTA Alert Minimum Terrain and Obstacle Clearance Values

Inhibited automatically (0,5 NM before and over the RWY when less than 200 ft above elevation) or manually, annunciation on PFD and MFD

TER INH



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

158

Compiled by Peter Schmidleitner

DIAMOND FLIGHT TRAINING



Traffic Avoidance TIS / TAS

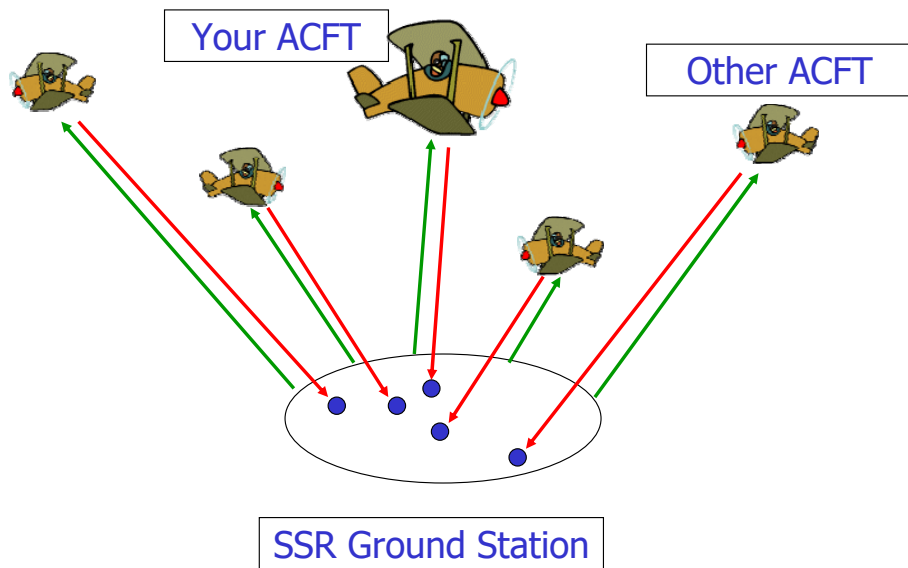


Traffic Advisory System

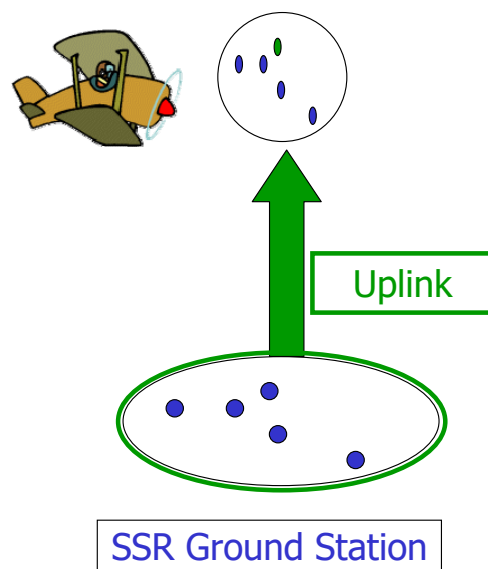
- TIS vs TAS:
 - TIS is a „passive (receiving) system“
 - TIS needs a TIS-equipped ground station
 - 5 sec update
 - Not available in Europe
 - TAS in an „active system“
 - TAS works independent from ground station
 - 0,5 sec update



Traffic Information System



Traffic Information System



Garmin G1000



Traffic Avoidance TAS



DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Traffic Advisory System



- Avidyne/Ryan 610
 - Max operating altitude: 25.000 ft



DIAMOND FLIGHT TRAINING

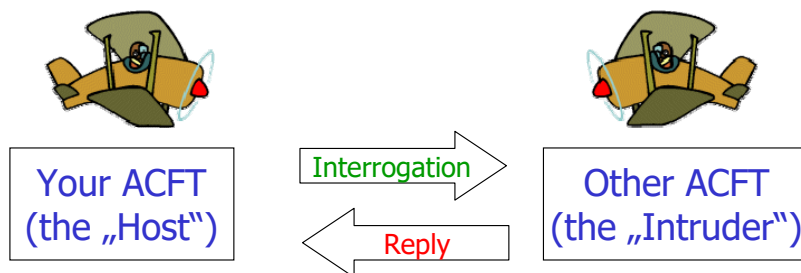
© Diamond Flight Training

164
Compiled by Peter Schmidleitner

Traffic Advisory System



TAS employs „active ranging“



Traffic Advisory System



- „Intruder“
 - a transponder equipped ACFT within the surveillance range for which the TAS has established a track
- „Traffic“
 - any nearby intruder
- Tau
 - the time to the intruder



Traffic Advisory System



- „Traffic“ may be:
 - **OT: Other Traffic**
 - an intruder within +/- 3500 ft and 12 NM that is not a PA or TA
 - **PA: Proximate Advisory**
 - an intruder within +/- 1200 ft and 5 NM
 - **TA: Traffic Advisory**
 - the nearest track of the intruder could result in a near-hit or a collision
 - **Scheduled Advisory**
 - a TA that is scheduled but not yet announced



Traffic Advisory System



- A **PA** (Proximate Advisory) will become a **TA** (Traffic Advisory) when:
 - the intruders **Tau** is less than the „TA threshold“
 - or
 - **range** and **altitude** separation are both less than the „TA threshold“



Traffic Advisory System



TA thresholds

Sensitivity Level A Gear DOWN	Intruder type	Host to Intruder		
		Tau (seconds)	Range (nm)	Altitude Separation (ft)
	Altitude reporting intruders	< 20	< 0.20	< 600
	Non altitude reporting intruders	< 15	< 0.20	N/A

Sensitivity Level B Gear UP	Intruder type	Host to Intruder		
		Tau (seconds)	Range (nm)	Altitude Separation (ft)
	Altitude reporting intruders	< 30	< 0.55	< 800
	Non altitude reporting intruders	< 25	< 0.20	N/A



Traffic Advisory System



- A **TA** (Traffic Advisory) remains in effect until:
 - range between host and intruder begins to diverge
 - or
 - intruder is no longer detected for 8 seconds
 - whichever is longer



Traffic Advisory System



- Non Threat Traffic—Hollow white diamond
- Proximity Advisories (PA)—Solid white diamond
- Traffic Advisories (TA)—Solid yellow circle
- Traffic Advisory Off Scale—Solid Split Circle, Yellow/Gray



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

171

Compiled by Peter Schmidleitner

Traffic Advisory System

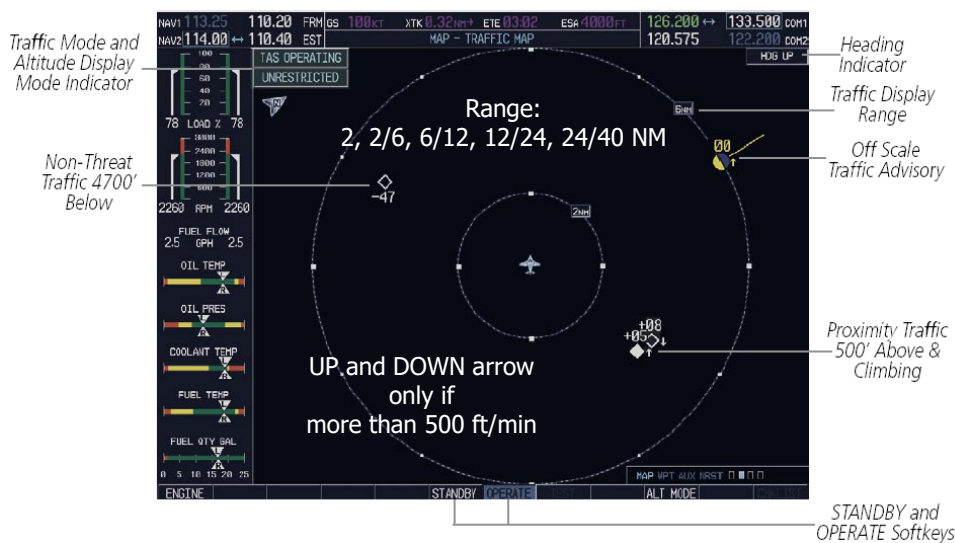


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

172

Compiled by Peter Schmidleitner

Traffic Advisory System



ENGINE STANDBY OPERATE TEST ALT MODE

Display of traffic:

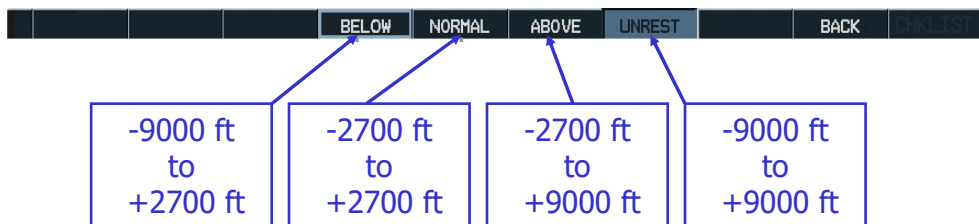


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

173

Compiled by Peter Schmidleitner

Traffic Advisory System

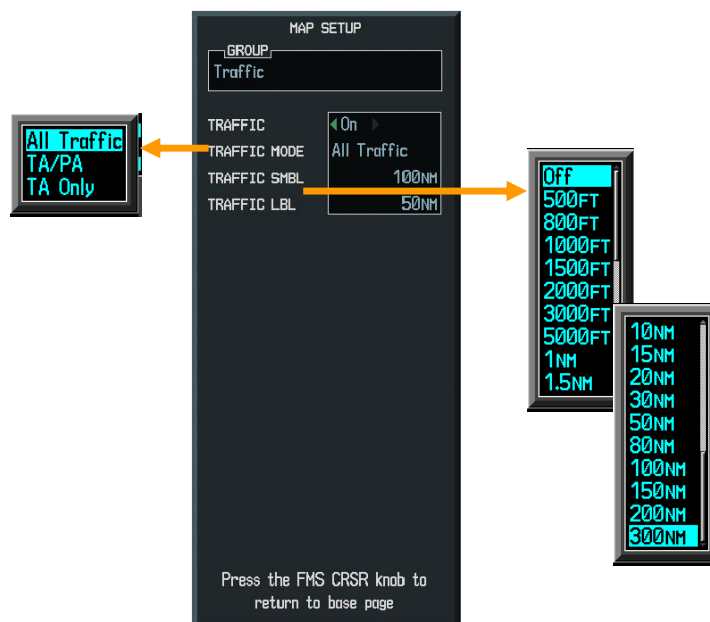


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

174

Compiled by Peter Schmidleitner

Traffic Advisory System



Audio Alerts

high

Traffic x o' clock same altitude x miles

low



Intentionally blank



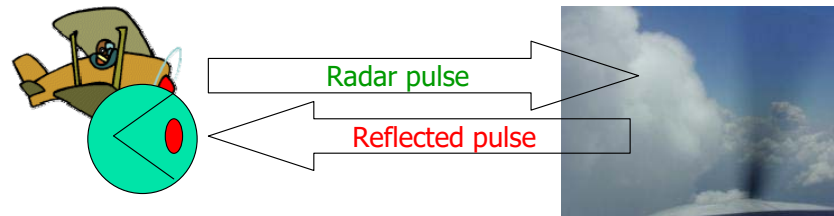
Garmin G1000



- Stormscope
- the poor man's weather radar ?



Weather Radar



Stormscope WX 500

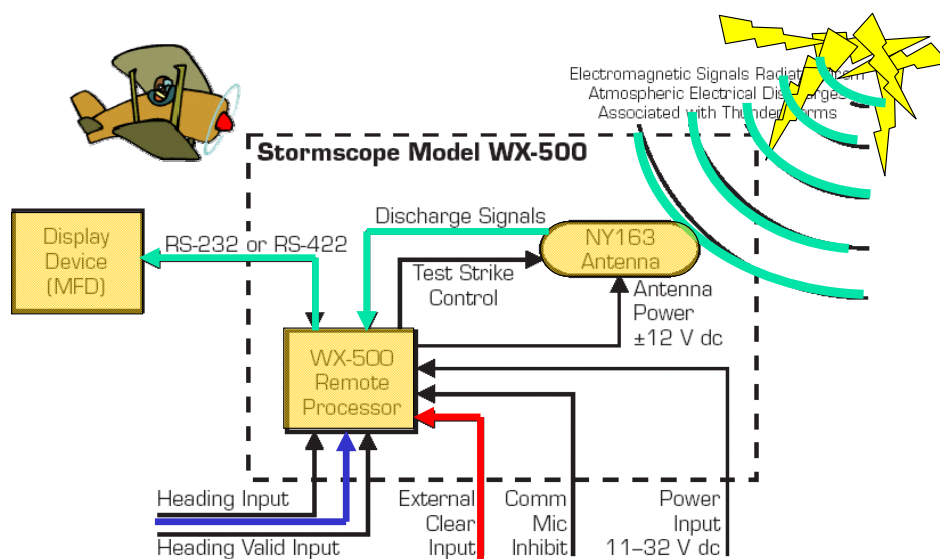


Figure 1-2. Functional Diagram



Stormscope WX 500

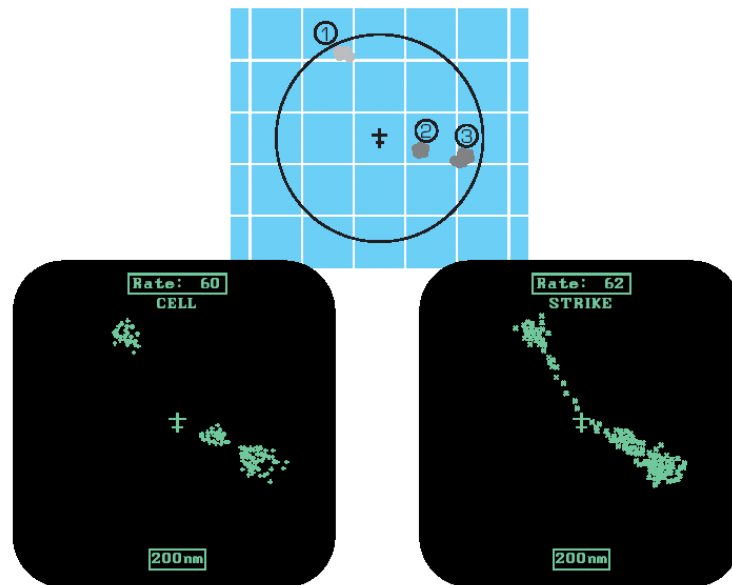


Figure 4-2. Three Clusters Within 200 nmi

Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

181

Compiled by Peter Schmidleitner

Stormscope WX 500



- Always operating
- Overlay on Map, or dedicated page with either 360 or 120 arc view modes
- Strike or Cell mode selected by soft key
- Range by use of Zoom/Pan joystick
- Data clear by soft key
- Strike aging
 - New strike is bold for 6 seconds
 - Normal lightning bolt for 1 minute
 - "+" 1 additional minute after strike
 - Smaller "+" for last minute



© Diamond Flight Training

182

Compiled by Peter Schmidleitner

Stormscope WX 500



Weather Map

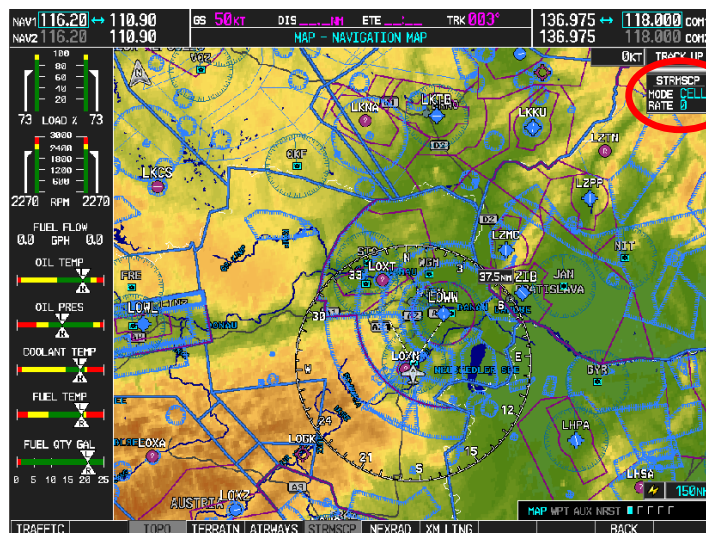


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

183

Compiled by Peter Schmidleitner

Stormscope WX 500



If map range < 25 NM
stormscope data are
not indicated.

This is announced
by:



Navigation Map



© Diamond Flight Training

184

Compiled by Peter Schmidleitner

GFC 700 AFCS „Autopilot“ GFC 700



GFC 700 AFCS

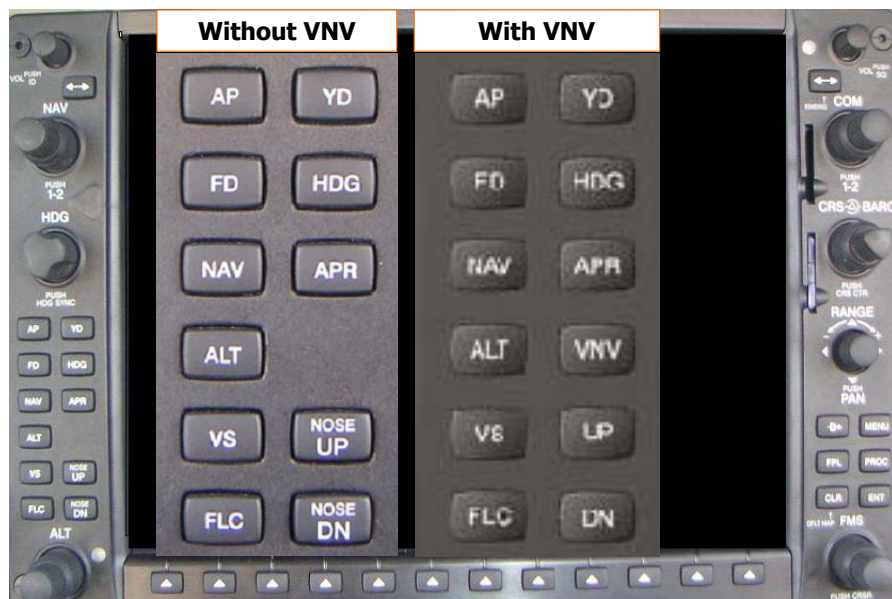
- „AFCS“:
 - Automatic Flight Control System
 - Flight Director
 - Autopilot



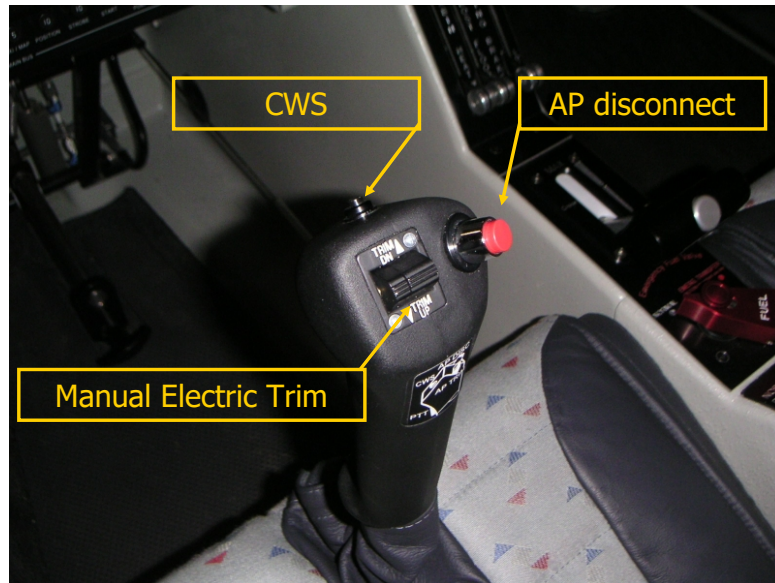
MFD Controls with GFC 700



MFD Controls with GFC 700



GFC 700 AFCS



GFC 700 AFCS



GFC 700 AFCS



- GFC 700 functions:
 - Flight Director (FD)
 - AUTO Pilot (AP)
 - Yaw Damper (YD)
 - Manual Electric Trim (MET)



GFC 700 AFCS



- Yaw Damper:
 - Dutch Roll damping
 - turn coordination

in response to

- yaw rate
- roll angle
- lateral acceleration
- airspeed

Yaw damper may remain operative in some cases when AP disengages

but:

DA42/62 Limitation:

The Yaw damper is an integral part of the autopilot system and must not be used separately



GFC 700 AFCS



Flight Director
Command Bars



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

193

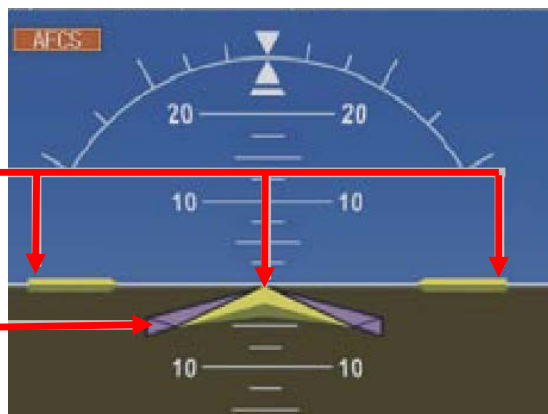
Compiled by Peter Schmidleitner

GFC 700 AFCS



Aircraft Attitude
Symbol

Flight Director
Command Bars



© Diamond Flight Training

194

Compiled by Peter Schmidleitner

GFC 700 AFCS

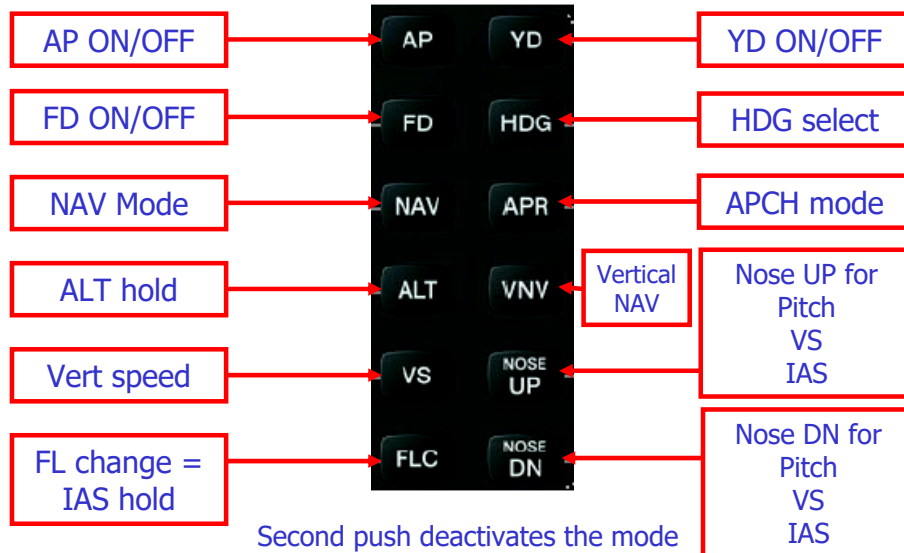


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

195

Compiled by Peter Schmidleitner

GFC 700 AFCS

- FLC („Flight Level Change“ Mode):
 - „FLC“ = IAS hold
 - A target altitude must be selected first
 - FLC does **not initiate** the level change to the selected altitude
 - Current altitude is maintained until either engine power or IAS reference are adjusted
 - FLC does not allow the aircraft to climb or descend away from the target altitude

© Diamond Flight Training

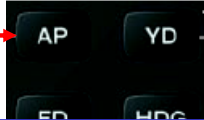
196

Compiled by Peter Schmidleitner

GFC 700 AFCS



AP ON/OFF



When the AP key is pressed, the autopilot, yaw damper and flight director are activated

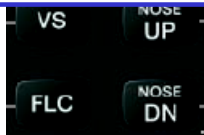


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

197

Compiled by Peter Schmidleitner

GFC 700 AFCS



When FD or AP is engaged:

Bank Angle	Flight Director Response
$< 6^\circ$	Rolls wings level
6° to 25°	Maintains current aircraft roll attitude
$> 25^\circ$	Limits bank to 25°



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

198

Compiled by Peter Schmidleitner

GFC 700 AFCS

System Status Annunciation

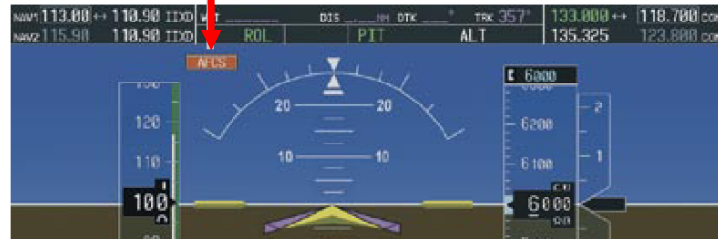


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

199

Compiled by Peter Schmidleitner

GFC 700 AFCS

Alert Condition	Annunciation	Description
Rudder Mistrim Right	RUD→	Yaw servo providing sustained force in the indicated direction
Rudder Mistrim Left	←RUD	
Aileron Mistrim Right	ATL→	Roll servo providing sustained force in indicated direction
Aileron Mistrim Left	←AIL	
Elevator Mistrim Down	↓ELE	Pitch servo providing sustained force in the indicated direction
Elevator Mistrim Up	↑ELE	
Pitch Trim Failure (or Stuck AP TRIM Switch)	PTRM	If AP engaged, take control of the aircraft and disengage AP If AP disengaged, move AP TRIM switches separately to unstick
Yaw Damper Failure	YAW	YD control failure; AP also inoperative
Roll Failure	ROLL	Roll axis control failure; AP inoperative
Pitch Failure	PTCH	Pitch axis control failure; AP inoperative
System Failure	AFCS	AP and MET are unavailable; FD may still be available
Preflight Test	PFT	Performing preflight system test; aural alert sounds at completion Do not press the AP DISC Switch during servo power-up and preflight system tests as this may cause the preflight system test to fail or never to start (if servos fail their power-up tests). Power must be cycled to the servos to remedy the situation.
	PFT	Preflight system test failed; aural alert sounds at failure



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

200

Compiled by Peter Schmidleitner

GFC 700 AFCS



Flight Mode Annunciators „AFCS Status Box“



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

201

Compiled by Peter Schmidleitner

GFC 700 AFCS



Flight Mode Annunciators „AFCS Status Box“



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

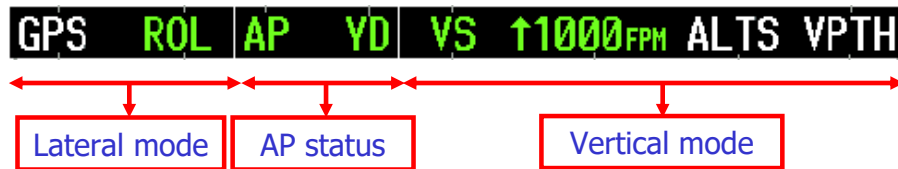
202

Compiled by Peter Schmidleitner

GFC 700 AFCS



Flight Mode Annunciators „AFCS Status Box“



Armed modes: **white**

Active modes: **green**



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

203

Compiled by Peter Schmidleitner

GFC 700 AFCS



Flight Mode Annunciators „AFCS Status Box“



Automatic disengagement

ROL	AP	YD	PIT	ALTS
ROL	AP	YD	PIT	ALTS

Manual disengagement

ROL	AP	YD	PIT	ALTS
-----	----	----	-----	------

Control Wheel Steering

ROL	CWS	YD	PIT	ALTS
-----	-----	----	-----	------



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

204

Compiled by Peter Schmidleitner

GFC 700 AFCS



Flight Mode Annunciators „AFCS Status Box“

GPS ROL AP YD VS ↑1000 FPM ALTS VPTH



Vertical mode



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

DIAMOND FLIGHT TRAINING

205

Compiled by Peter Schmidleitner

GFC 700 AFCS



Vertical modes

Vertical Mode	Control	Annunciation
Pitch Hold	(default)	PIT
Selected Altitude Capture	*	ALTS
Altitude Hold	ALT Key	ALT nnnnn FT
Vertical Speed	VS Key	VS nnnn FPM
Flight Level Change	FLC Key	FLC nnn KT
Vertical Path Tracking	VNV Key	VPTH
VNV Target Altitude Capture	**	ALTV
Glidepath***	APR Key	GP
Glideslope		GS
Go Around	GA Button	GA



© Diamond Flight Training

DIAMOND FLIGHT TRAINING

206

Compiled by Peter Schmidleitner

GFC 700 AFCS



GA mode active

GA **AP YD** **GA** **ALTS**

- Wings level
- 6°_(DA42) 7°_(DA40) pitch up



GFC 700 AFCS

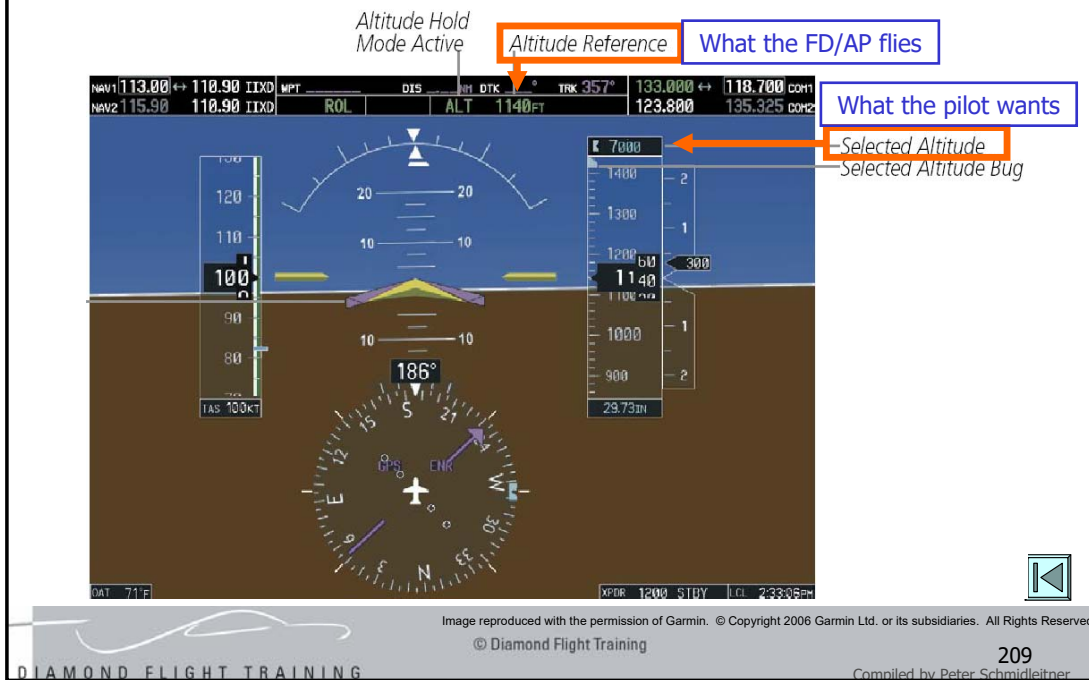


Vertical mode range and change increments

Pitch Mode	Control	Annunciation	Reference Range	Reference Change Increment
Pitch Hold	(default)	PIT	-15° to +20°	0.5°
Altitude Hold	ALT Key	ALT nnnnn FT		
Vertical Speed	VS Key	VS nnnn FPM	-3000 to +1500 fpm	100 fpm
Flight Level Change	FLC Key	FLC nnn KT	80 to 180 kt	1 kt
Glideslope Arm/Capture/Track	APR Key	GS		
Go Around	GA Button	GA		



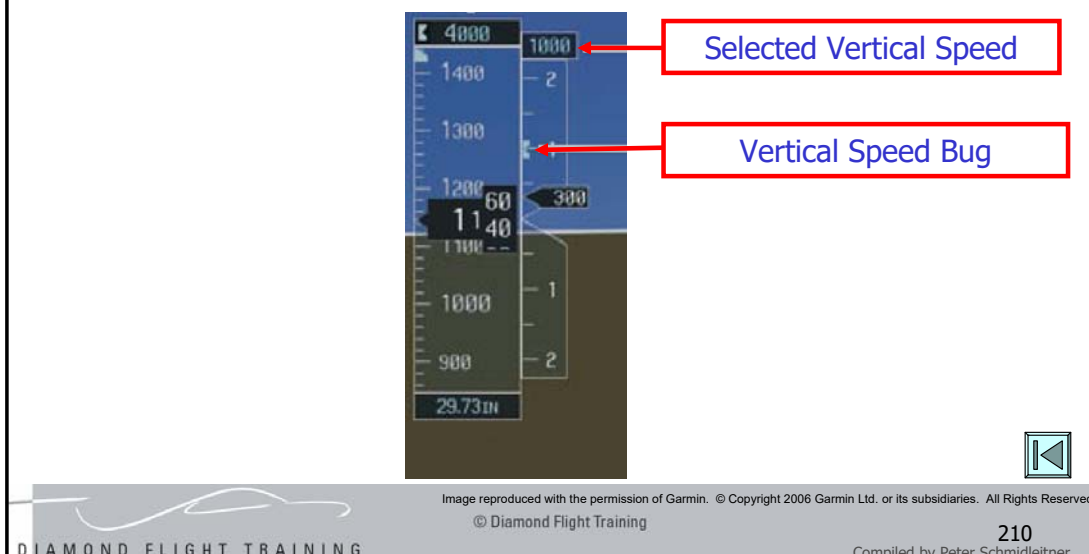
GFC 700 AFCS



GFC 700 AFCS



Vertical Speed Mode



GFC 700 AFCS



IAS Hold Mode („Flight Level Change“)

Selected Airspeed

Airspeed Reference Bug



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

211

Compiled by Peter Schmidleitner

GFC 700 AFCS



Overspeed Protection



flashing

Pitch commands are limited so that the selected vertical mode does not exceed the maximum autopilot airspeed



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

212

Compiled by Peter Schmidleitner

GFC 700 AFCS



Flight Mode Annunciators „AFCS Status Box“



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

213

Compiled by Peter Schmidleitner

GFC 700 AFCS



Roll Mode	Control	Annunciation
Roll Hold	(default)	ROL
Heading Select	HDG Key	HDG
Navigation, GPS Arm/Capture/Track	NAV Key	GPS
Navigation, VOR Enroute Arm/Capture/Track		VOR
Navigation, LOC Arm/Capture/Track (No Glideslope)		LOC
Navigation, Backcourse Arm/Capture/Track		BC
Approach, GPS Arm/Capture/Track	APR Key	GPS
Approach, VOR Arm/Capture/Track		VAPP
Approach, ILS Arm/Capture/Track (Glideslope Mode automatically armed)		LOC
Go Around	GA Button	GA



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

214

Compiled by Peter Schmidleitner

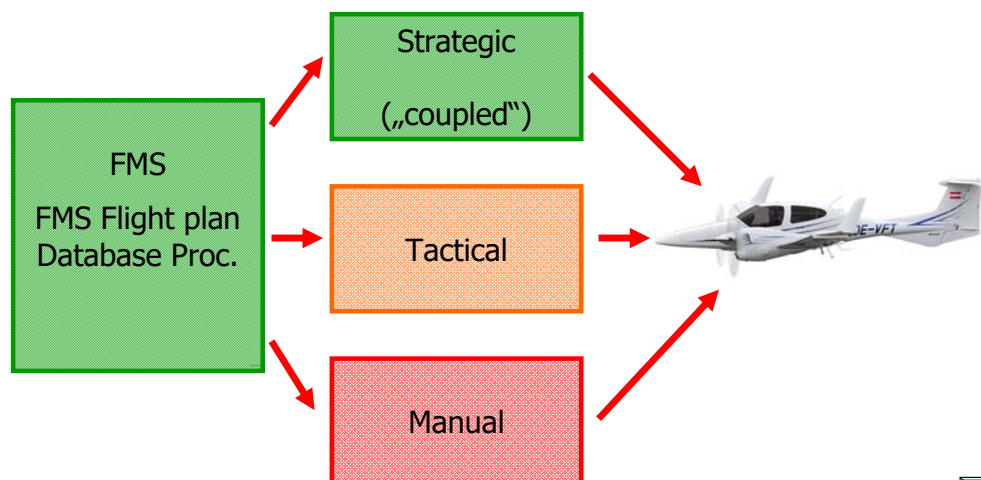
Garmin G1000

Never forget ...



FMS – ACFT „connection“

„Connection“ to ACFT



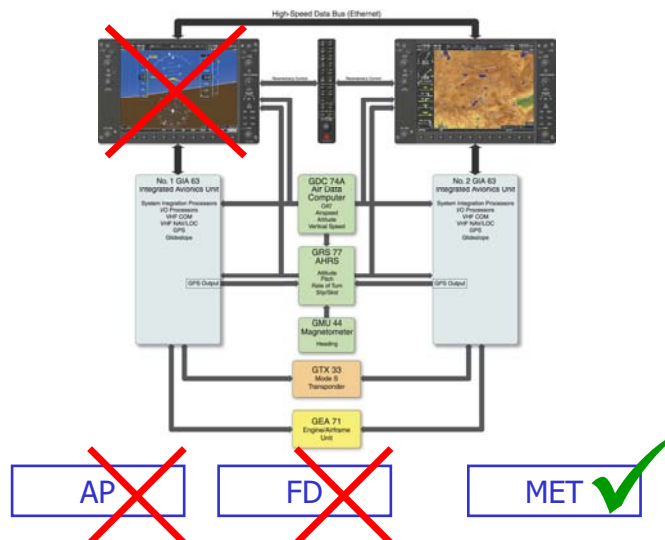
G 1000 System Loss Effect upon Autopilot Operation

AHRS	The Autopilot disconnects and Autopilot, Yaw Damper and Flight Director are inoperative. Manual electric trim is available.
HDG function of AHRS	The Autopilot will remain engaged with the loss of the HDG Mode.
MFD	The Autopilot will remain engaged with limited functionality.
PFD	The Autopilot disconnects and Autopilot and Flight Director are inoperative. Manual electric trim is available.
GIA No. 1	The Autopilot disconnects and Autopilot, Flight Director and manual electric trim are inoperative.
GIA No. 2	The Autopilot disconnects and Autopilot and manual electric trim are inoperative. Flight Director is available.
GPS No. 1 and 2	The Autopilot and Flight Director operates in NAV modes only (LOC, BC, VOR, VAPP) with reduced accuracy.
ADC	The Autopilot disconnects and Autopilot is inoperative. The flight director is available except for air data modes (ALT, VS, FLC). Manual electric trim is available.

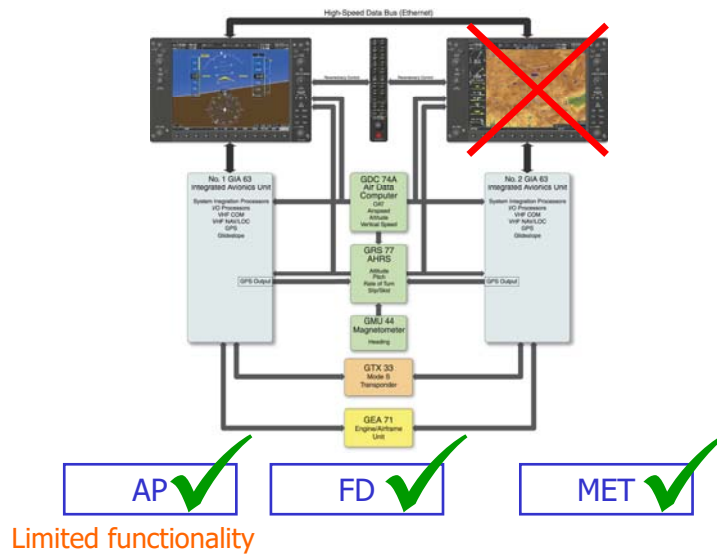
The yaw damper can remain operational in some cases where the autopilot automatically disengages.



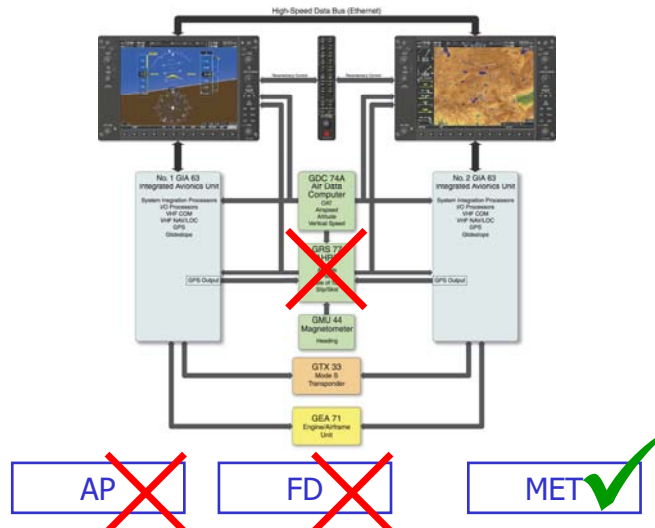
G 1000 System Loss Effect upon Autopilot Operation



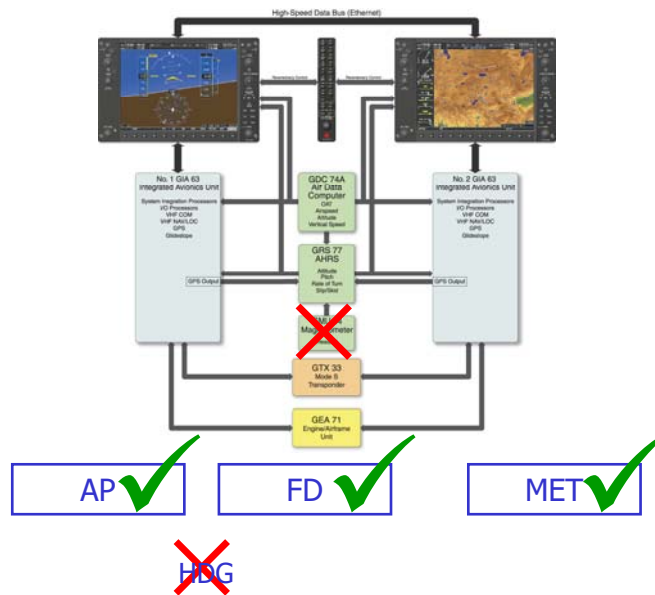
G 1000 System Loss Effect upon Autopilot Operation



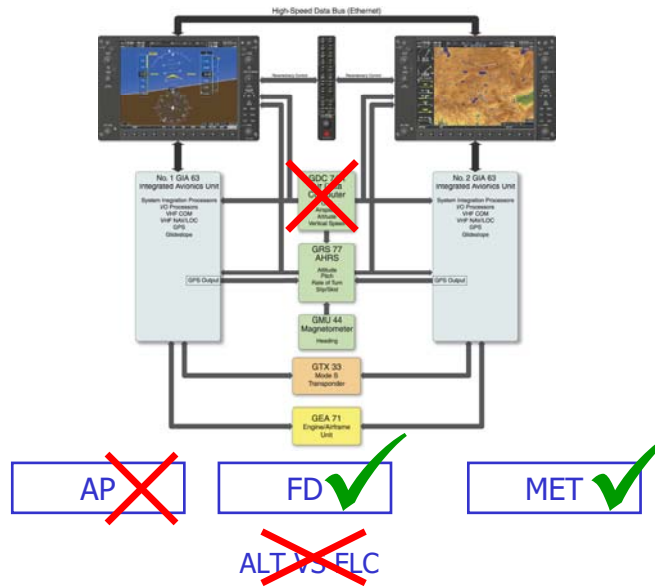
G 1000 System Loss Effect upon Autopilot Operation



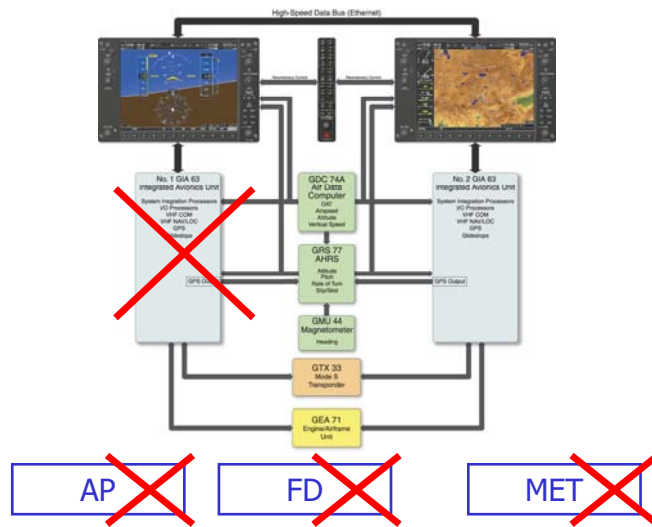
G 1000 System Loss Effect upon Autopilot Operation



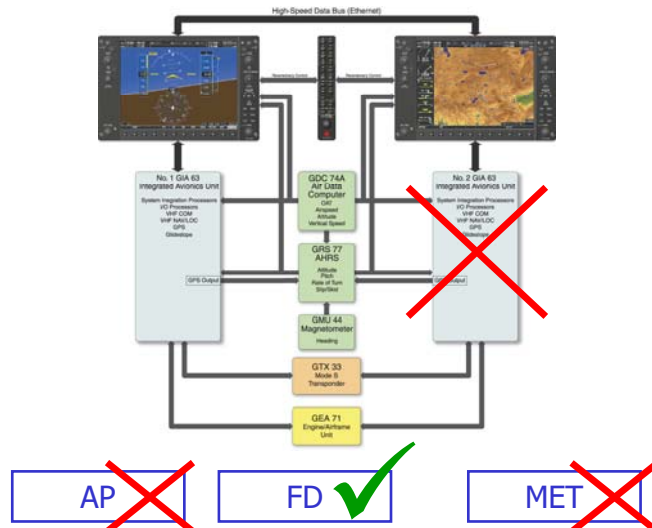
G 1000 System Loss Effect upon Autopilot Operation



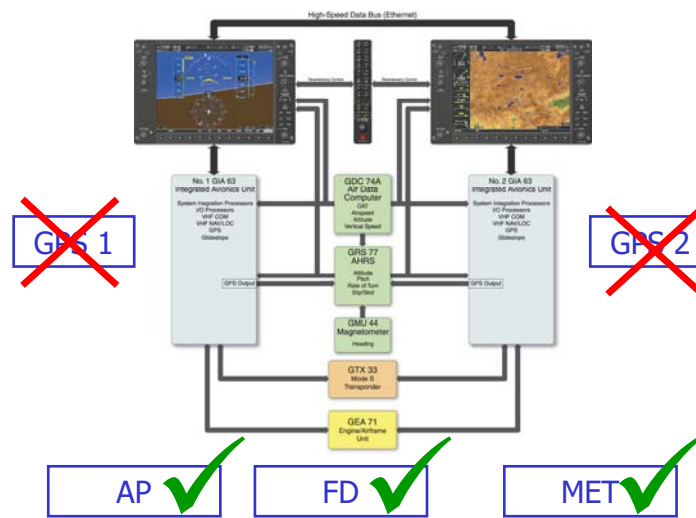
G 1000 System Loss Effect upon Autopilot Operation



G 1000 System Loss Effect upon Autopilot Operation



G 1000 System Loss Effect upon Autopilot Operation



VHF NAV only, reduced accuracy



Intentionally blank



Garmin G1000



„Safe Taxi“ and „Jeppesen Chart View“



Limitation



The availability of SafeTaxi®, ChartView, or FliteCharts® in electronic form on the G1000 is for information purposes only, it is still mandatory to carry another source of charts on-board the airplane.



„SafeTaxi“



Taxiway
Identification

Aircraft
Position

Airport Hot
Spot Outline

Airport
Features

DCLTR Softkey
Removes Taxiway
Markings



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

229

Compiled by Peter Schmidleitner

Jeppesen „Chart View“

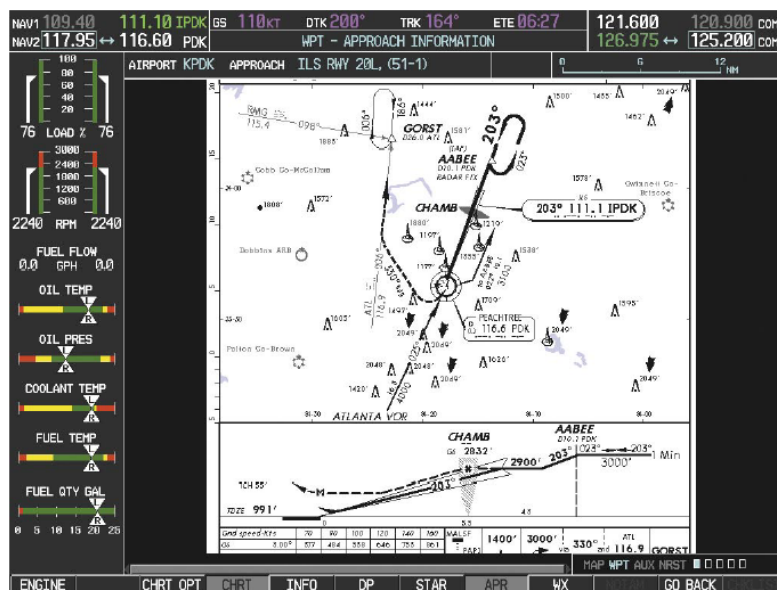


Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

230

Compiled by Peter Schmidleitner

Jeppesen „Chart View“



Power-up Splash Screen

DATABASE	
✓ Checklist File: N/A	
🛣 Basemap Land 2.00	
➡ SafeTaxi Expires 20-DEC-2007	
🏔 Terrain 2.04	
✈ Airport Terrain 2.04	
⚡ Obstacle Expires 20-DEC-2007	
✈ Aviation Expires 22-NOV-2007	
🗺 ChartView Disables 24-JAN-2008	
All map and terrain data provided is only to be used as a general reference to your surrounding and as an aid to situational awareness.	

AUX - System Status

DATABASE	
VERSION	2.08
CYCLE	07B6
EFFECTIVE	25-OCT-07
EXPIRES	20-DEC-07
Copyright 2007 Garmin Ltd. or its subsidiaries	
CHART	
CHART	Jeppesen ChartView
REGION	NOT AVAILABLE
CYCLE	NOT AVAILABLE
EFFECTIVE	-- -- --
EXPIRES	-- -- --
DISABLES	-- -- --



Image reproduced with the permission of Garmin. © Copyright 2006 Garmin Ltd. or its subsidiaries. All Rights Reserved
© Diamond Flight Training

231

Compiled by Peter Schmidleitner

NXi



■ Dual chart type support:

- Simultaneous use of AeroNav FliteCharts and Jeppesen ChartView supported



© Diamond Flight Training

232

Compiled by Peter Schmidleitner

Garmin G1000



DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

On RWY 31 in LOAV
without Synthetic Vision



DIAMOND FLIGHT TRAINING

© Diamond Flight Training

234

Compiled by Peter Schmidleitner

On RWY 31 in LOAV with Synthetic Vision

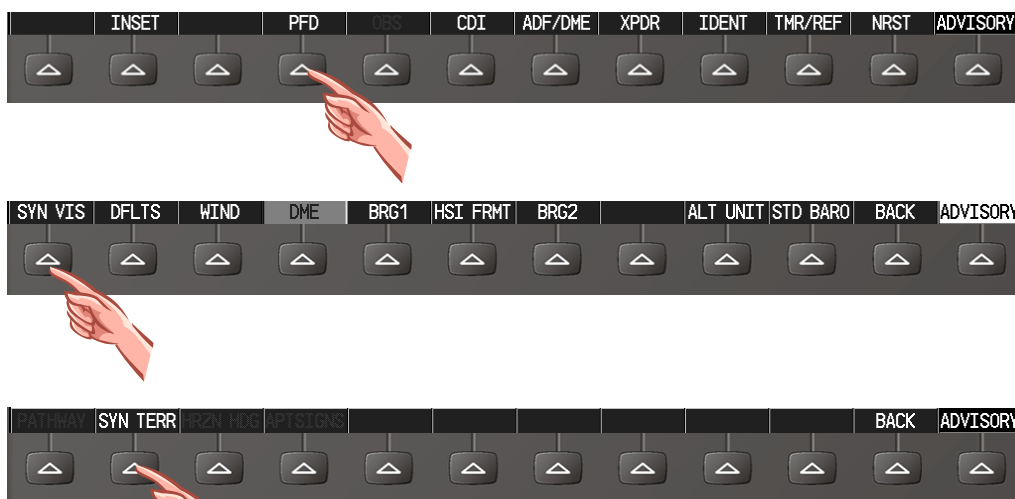


© Diamond Flight Training

235

Compiled by Peter Schmidleitner

How to get Synthetic Vision



© Diamond Flight Training

236

Compiled by Peter Schmidleitner

Diamond
FLIGHT TRAINING

237

Compiled by Peter Schmidleitner

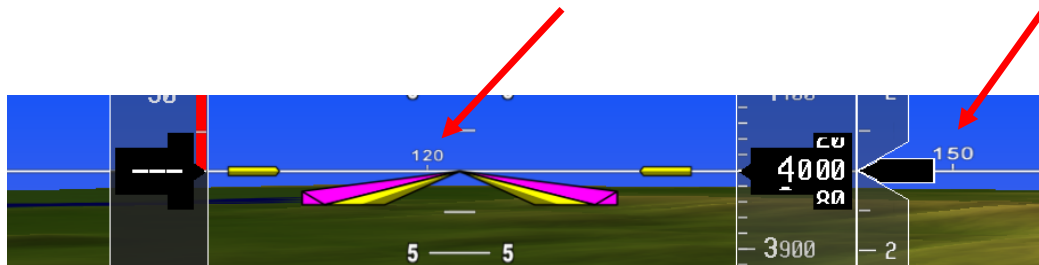
Diamond
FLIGHT TRAINING



238

230
Compiled by Peter Schmidleitner

Horizon Heading



DIAMOND FLIGHT TRAINING

© Diamond Flight Training

239

Compiled by Peter Schmidleitner

APT Signs



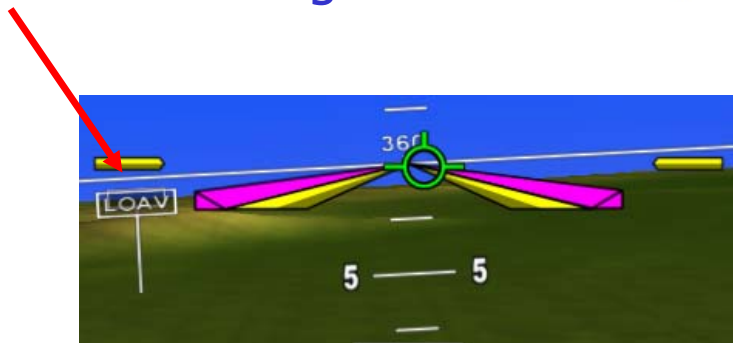
DIAMOND FLIGHT TRAINING

© Diamond Flight Training

240

Compiled by Peter Schmidleitner

APT Signs



APT signs visible within 15 NM

Text visible within 8 NM

Text removed within 4,5 NM, only RWY is visible

RWY picture is proportional at $\sim 1/2$ NM



Pathway



Pathway



© Diamond Flight Training

243

Compiled by Peter Schmidleitner

Pathway



© Diamond Flight Training

244

Compiled by Peter Schmidleitner

Pathway



© Diamond Flight Training

245

Compiled by Peter Schmidleitner

Pathway



© Diamond Flight Training

246

Compiled by Peter Schmidleitner

Pathway



© Diamond Flight Training

247

Compiled by Peter Schmidleitner

Pathway



© Diamond Flight Training

248

Compiled by Peter Schmidleitner

Pathway



© Diamond Flight Training

249

Compiled by Peter Schmidleitner

Pathway



© Diamond Flight Training

250

Compiled by Peter Schmidleitner

Pathway



© Diamond Flight Training

251

Compiled by Peter Schmidleitner

Intentionally blank

© Diamond Flight Training

252

Compiled by Peter Schmidleitner

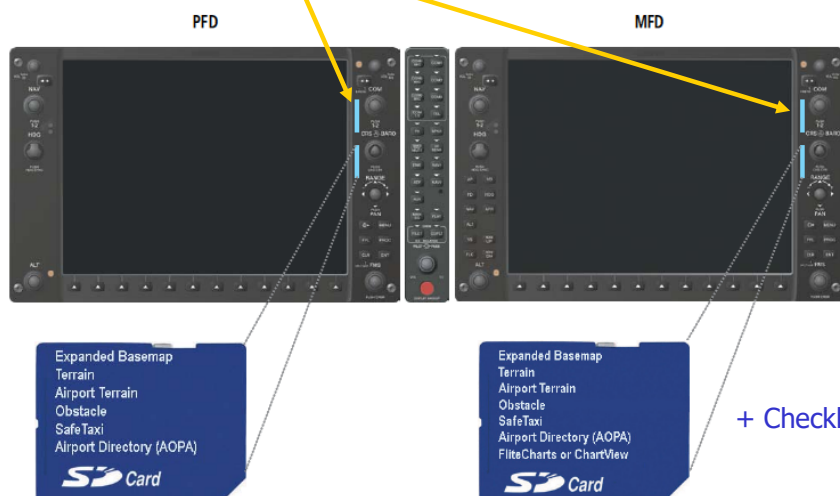
Garmin G1000



Databases



Load Jeppesen Navigation Database here



■ Database management:

- A standby database can be stored and standby / active can be swapped



Intentionally blank



Garmin G1000



Additional Features



DIAMOND FLIGHT TRAINING

© Diamond Flight Training

Compiled by Peter Schmidleitner

Additional Features



- Electronic Checklist
- Auxiliary Video (Optional)
- Flight Data Logging



DIAMOND FLIGHT TRAINING

© Diamond Flight Training

258
Compiled by Peter Schmidleitner

Electronic Checklist



© Diamond Flight Training

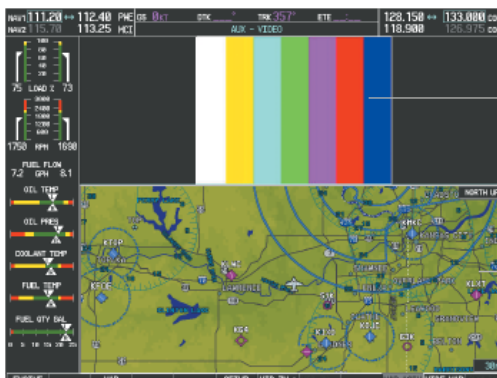
259

Compiled by Peter Schmidleitner

Auxiliary Video

Full screen

Split screen



© Diamond Flight Training

260

Compiled by Peter Schmidleitner

Flight Data Logging



- SD card in upper slot of MFD
- ~4000 flight hours logged on card
- Data logged once each second when MFD is powered
- For data logging status see AUX-UTILITY page
- Data stored in one file per day
- Log file format:
log_110525_153024_LOAN.csv
yyymmdd_hhmmss_nearest APT



Thank you for your attention!

